CoMem: Reusing Transformer Depth across Queries with Persistent Intermediate Residuals

Hanzuo Liu[†‡] **Xuan Qi**[†] **Chunyu Liu**[†] **Haotian Zhong**[†]
Yulong Wang[‡] **Key**[‡] **Rayying**[‡] **Alex Lamb**[†] **Mingyu Gao**[†]
[†]Tsinghua University [‡]Tencent
{lhz24@mails., gaomy@}tsinghua.edu.cn
https://github.com/liuhanzuo/Mixture-of-Memory

Abstract

Repeated queries over shared documents repeatedly execute the same lower transformer layers. We introduce **CoMem**, which makes split depth j an explicit reusable-context axis: write one depth-j residual per token, select a bounded chunk set, and resume only layers $[j{:}L)$. Among document-reuse systems we are aware of, CoMem jointly makes split depth a tunable serving axis and isolates it with a matched $j{=}0$ endpoint. On Qwen3-8B, $j{=}12$ reduces selected-pack Read from 931.9 to 664.4 ms (1.403×), with a 3.12-point RULER cost (95% CI $[2.36, 3.93]$); a continuous-prefix oracle recovers the full gap. The resulting depth axis quantifies a quality–latency–storage trade-off; a separate same-adapter, Write-inclusive pipeline is 2.74× faster. Equal-latency raw replay leads by 11.56 points with BM25, directly measuring an applicability boundary of prepaid depth rather than hiding it. CoMem stores 8 KiB/token versus 144 KiB/token for a protocol-aligned same-Qwen3 CacheBlend-style diagnostic; the cohorts and adaptation budgets are not matched. A context–position factorization identifies missing lower-layer document context as the dominant tested multikey error, and a 32-token overlap raises 92.5 to 98.5 without increasing persistent bytes or per-query Read. CoMem opens transformer depth as a measurable, tunable dimension for repeated-query long-context serving.

1 Introduction

Long-context applications repeatedly ask new questions about the same manuals, repositories, histories, or document collections. Raw-text retrieval reduces the number of online tokens, but every selected token still traverses every transformer layer (Lewis et al., 2020; Xu et al., 2024). Reusable-context systems move this work offline: CacheBlend, TurboRAG, EPIC, APE, and related systems cache layer-wise KV and repair context or position dependencies (Yao et al., 2025; Lu et al., 2025; Hu et al., 2025; Yang et al., 2025b; Agarwal et al., 2025). Yet these interfaces largely fix what is stored and how much model depth has already been executed. This leaves a basic systems question unanswered: *how much transformer depth should each reusable document object prepay?*

We answer this question with **CoMem**, which exposes the split j between offline document processing and online query processing. Write runs each chunk through layers $[0{:}j)$ and stores its residual h_j. Select retrieves top-k chunks. Read packs the selected residuals with the query and executes only $[j{:}L)$ with full cross-pack attention. The $j{=}0$ endpoint stores token IDs and replays the identical pack through all layers. Varying j therefore trades persistent bytes and write-once computation against per-query model depth while holding retrieved evidence fixed.

Prior activation-reuse methods cache a chosen intermediate representation for repeated downstream runs (Lin et al., 2021; Saad-Falcon et al., 2023); LLMCache maintains per-layer activation banks and semantically matches similar inputs at arbitrary layers (Bansal, 2025); HCache stores hidden states across layers to reconstruct layer-wise KV (Gao et al., 2025); and

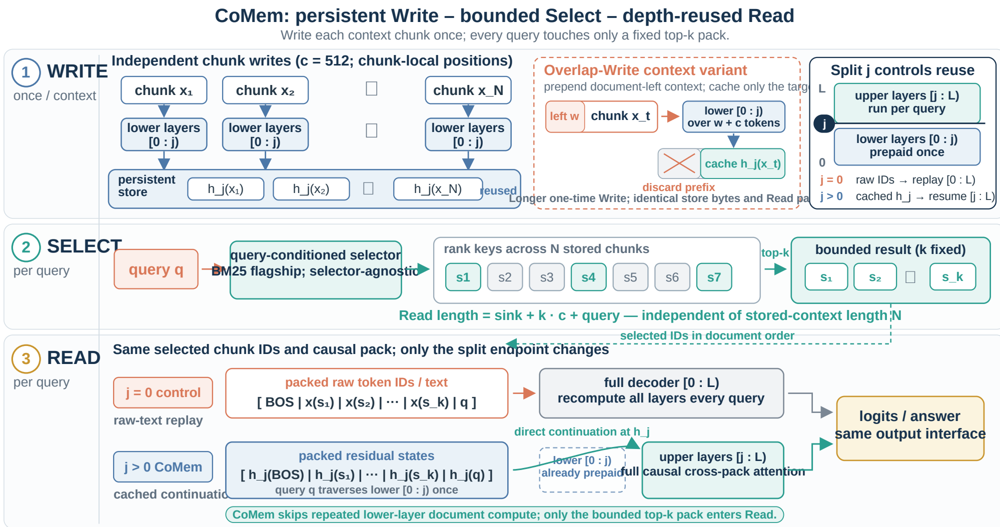

Figure 1: **CoMem turns transformer depth into a cross-query reuse axis.** WRITE prepays layers $[0{:}j)$ once per document chunk, SELECT retrieves a bounded working set, and READ resumes $[j{:}L)$ with the query present. The matched $j{=}0$ endpoint uses the same selected chunks, order, sink, mask, and adapter while replaying all layers, thereby isolating the incremental depth effect. Overlap-Write supplies a small amount of left document context without enlarging the stored object or Read pack.

PIC/modular-cache systems store or learn layer-wise KV objects (Yao et al., 2025; Hu et al., 2025; Chen et al., 2026; Eyuboglu et al., 2026). CoMem's specific conjunction is a persistent document object at one chosen split, direct decoder-suffix continuation, and an identical-evidence $j{=}0$ endpoint. This formulation makes prepaid depth directly measurable rather than an implementation constant.

The matched result establishes a clear quality–latency trade-off. On Qwen3-8B, moving from $j{=}0$ to $j{=}12$ reduces selected-pack Read from 931.9 to 664.4 ms, a 1.403× speedup, while the paired 15-cell RULER macro changes from 99.19 to 96.07 (gap 3.12, 95% CI [2.36, 3.93]). A rank-32 self-distillation adapter makes the split interface effective while leaving the backbone frozen. When bounded selection and depth reuse are combined, a separate H20 Pareto harness reduces store-ready 128k online prefill from 50.59 to 1.32 s. This operating point compares stock, LoRA-off dense prefill with adapted CoMem and excludes one-time Write and external fetch; it is therefore not a same-adapter estimate. A different same-adapter, Write-inclusive L20A pipeline improves from 6.035 to 2.202 s (2.74×). The 1.403× result isolates prepaid depth; the other two measure composed deployment paths under explicitly different boundaries.

CoMem further provides a unified measurement of a single-residual reusable object across quality, per-query latency, persistent storage, one-time Write, and empirical break-even. The residual costs 8 KiB/token—18× smaller than full bf16 KV for Qwen3-8B, but much larger than text. In the write-once serving harness, CPU-pinned storage breaks even after 8.9–10.9 repeated queries at 32k for generations up to 128 tokens; at 128k and one generated token, GPU/CPU placement breaks even after 25.8–27.6 queries. The full serving grid extends these measurements to 1M stored tokens and four generation lengths (Table 5).

The experiments yield two further scientific findings. First, a continuous-prefix h_{12} oracle exactly recovers the 3.12-point matched gap, locating the attainable loss at the reusable Write interface rather than the capacity of the upper decoder. A 2×2 context–position factorization then identifies missing lower-layer document context as the dominant tested factor on paired 8k/16k multikey instances: full document context raises 92.5 to 100.0, while remapping positions alone lowers it to 88.0. A deployable 32-token overlap reaches 98.5

Family	**Stored state**	**Tunable axis**	**Online work**
Text replay	token IDs	selection budget	all layers
Activation reuse	intermediate / multi-layer states	matching / restoration	downstream or rebuilt layers
Full-depth KV reuse	cached or learned KV	linking / repair	repair + decode
CoMem	one residual / token	**split depth** j	suffix $[j{:}L)$

Table 1: **Compact positioning of reusable-context interfaces.** CoMem makes prepaid depth tunable and provides a matched $j{=}0$ endpoint; detailed distinctions appear in Appendix A.1.

with unchanged persistent bytes and per-query Read. Second, the equal-latency comparison delineates where depth reuse applies. Under the common iterative-BM25 ranking, raw replay leads CoMem by 11.56 points: spending the saved latency on full-depth replay is preferable for this tested mixture. When only replay uses frozen BGE, its 1.00-point edge is unresolved under hierarchical cell resampling. These statistical boundaries make retrieval policy a first-order coordinate of the deployment frontier.

Scope. Our central claims concern the matched depth axis, measured amortization, and the tested Write-context mechanism. CoMem targets repeated queries over stable corpora with open-weight models; the nearest PIC and learned modular-KV systems use different backbones and measurement boundaries. We therefore pair the broader conceptual comparison with one same-Qwen3 minimal-faithful CacheBlend-style experiment rather than transfer unmatched published numbers. The full set of deployment, generalization, and statistical limits is consolidated in the dedicated Limitations section.

Contributions.

- **A depth-partitioned memory abstraction.** CoMem combines independently writable intermediate residuals, bounded content selection, and direct upper-layer replay, with split depth jointly controlling storage, fidelity, and query-time work.
- **A matched $j{=}0$ endpoint design.** Identical-pack replay cleanly separates the depth-reuse effect from retrieval and packing; oracle and factorized controls localize the deployable fidelity gap to independent chunk writing.
- **A quantified quality–storage–latency frontier.** We measure depth splits, selectors, and deployment scenarios, including a qualified same-backbone prior-art comparison and an equal-latency boundary that defines when depth reuse is and is not beneficial.

2 Related Work

We organize prior work by the persistent object and the model computation that remains after retrieval (Table 1). This comparison exposes the specific conjunction studied by CoMem: a persistent document residual at one chosen split, direct decoder-suffix execution, and an identical-evidence $j{=}0$ endpoint for measuring the incremental depth effect.

Reusable intermediate representations. ReadOnce Transformers construct compressed, task-independent document representations for repeated downstream use, while Embedding Recycling caches an intermediate encoder layer and trains adapters above it (Lin et al., 2021; Saad-Falcon et al., 2023). These works establish that intermediate text representations can be persistent and useful, but their main trade-off is representation compression or downstream adaptation; the cached layer is a chosen architecture setting rather than a measured decoder-serving axis. CoMem operates in an autoregressive decoder, keeps one residual per selected token, and varies the suffix depth executed for each query.

LLMCache caches intermediate activations in an independent bank for every transformer layer and uses semantic fingerprints to reuse a matching state for similar inputs (Bansal, 2025). It therefore makes cache behavior layer-dependent and supports arbitrary layers, but it does not define one document residual split followed by a fixed native suffix, nor an identical-evidence $j=0$ endpoint. HCache and KV-Direct are closer at the state level. HCache saves hidden states throughout the model and reconstructs each layer's KV for fast restoration; KV-Direct reconstructs layer-wise KV from residual checkpoints (Gao et al., 2025; Qasim et al., 2026). Their goal is efficient restoration of the standard layer-wise KV state. CoMem chooses one split residual and executes the remaining transformer blocks directly, which creates a different storage–depth–quality trade-off and admits the matched $j=0$ control.

Full-depth chunk-KV reuse. Prompt Cache reuses exact-match prompt modules, while Block-Attention and TurboRAG independently precompute chunk KV and restore positional consistency when modules are assembled (Gim et al., 2024; Ma et al., 2025; Lu et al., 2025). CacheBlend adds selective recomputation to repair cross-chunk dependencies; RAGCache and Cache-Craft optimize management and repair around the same full-depth object (Yao et al., 2025; Jin et al., 2025; Agarwal et al., 2025). These systems avoid most document recomputation by storing KV at every layer. CoMem instead makes *cached depth* the adjustable resource: one h_j residual replaces the full-depth KV object, and layers $[j{:}L)$ reconstruct query-conditioned state. Our same-Qwen3 comparison (Table 6) makes this difference empirical: a minimal faithful CacheBlend-style implementation with global RoPE reindexing and up to 18% selective recomputation reaches 67.80–74.70 RULER, 48.50–49.83 BABILong qa5, and 16.77–17.37 LoCoMo substring accuracy, versus CoMem's 97.05, 68.67, and 23.36 while storing 18$\times$ fewer bytes/token. This comparison isolates a full-depth chunk-KV alternative under one backbone and evidence pack; it does not replace each system's native serving implementation.

PIC and learned modular document caches. EPIC and MEPIC formalize position-independent caching and reduce link-time recomputation; APE encodes contexts in parallel and realigns attention (Hu et al., 2025; Wang et al., 2025; Yang et al., 2025b). Fusion RAG Cache further studies cache selection and fusion at serving scale (Wang et al., 2026). Learned modular objects include KV Packet's compiled context-independent KV and Cartridges' self-studied reusable KV representations (Chen et al., 2026; Eyuboglu et al., 2026). Cartridges at Scale and SemPIC extend learned per-document KV retrieval (Hardalov et al., 2026; Xie et al., 2026). They optimize how a full-depth KV object is compiled, linked, or retrieved; CoMem exposes a single-residual depth axis and quantifies the compute that remains after selection.

Token selection, compression, and external memory. ILRe, REFORM, and GemFilter use intermediate layers to select or gather tokens before later computation, but do not persist the selected hidden states across queries (Liang et al., 2025; Song et al., 2025; Shi et al., 2026). KV-compression methods reduce retained state after processing (Xiao et al., 2024b; Zhang et al., 2023; Li et al., 2024; Cai et al., 2025; Liu et al., 2024). MemoryLLM, LongMem, and XC-Cache add latent pools or auxiliary readers (Wang et al., 2024; 2023; Monteiro et al., 2024). CoMem addresses a separate repeated-query boundary: prepay a chosen number of native decoder layers once, retrieve a bounded residual pack, and resume the unmodified suffix through a small adapter.

Positioning. The nearest systems differ in backbone, training, tasks, storage tier, and timing boundary, precluding a numerical cross-paper ranking. The defensible novelty is narrower and structural: CoMem makes transformer split depth the controlled variable of reusable-context serving, provides an identical-pack $j=0$ endpoint, and measures the resulting residual object across quality, latency, storage, Write amortization, and context dependence.

3 Why Reuse Model Depth?

The interface does not require a layer at which a model has "understood" the document. It requires only that intermediate states retain useful information and that later layers can be

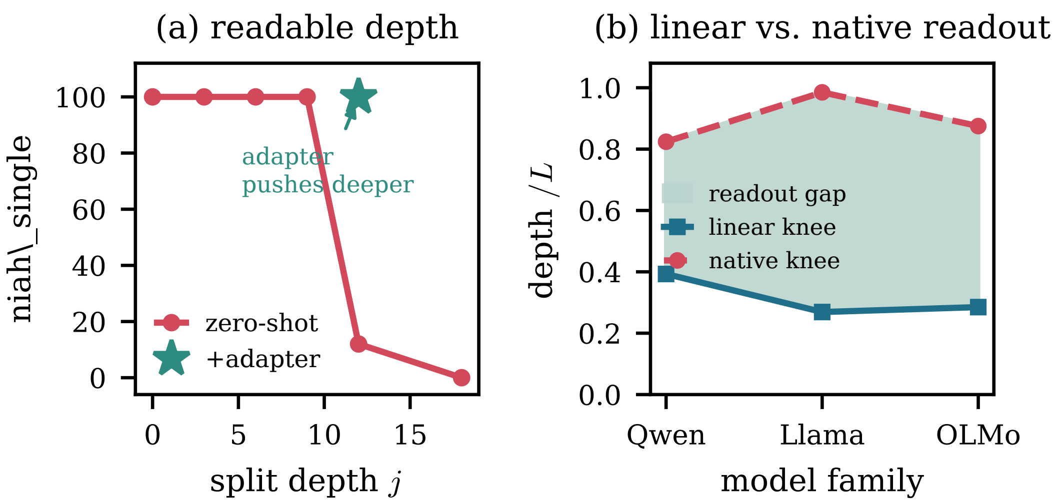

Figure 2: **Intermediate accessibility motivates, but does not validate, depth reuse.** Frozen probes become informative before native LM-head readout, but both measures are protocol-dependent; definitions and controls are in Appendix A.2.

adapted to read states produced without the query. Figure 2 supplies motivation, while the matched experiment in Section 5 supplies the actual evidence.

A frozen interface deteriorates as the split moves deeper. With separately distilled adapters, RULER changes from 98.29 at j=6 to 96.07 at j=12 and 55.41 at j=18, while fixed-pack Read speedup rises from 1.166× to 1.403× and 1.807× (Appendix Table 12). Each split uses a separately distilled suffix adapter, yielding a practical *multi-depth deployment curve*: greater prepaid depth systematically reduces Read time until interface quality collapses at j=18. Because adapter spans and parameter counts change with j and the retained j=12 Write value is unavailable, the clean causal estimate remains the matched j=0 versus j=12 endpoint.

Selection and representation also fail separately. BM25 retains lexical single-needle evidence, but on LongEval the distilled j=12 reader reaches only 64–76% accuracy conditional on a retrieval hit; BABILong exhibits both misses and hit-conditional readout failures. We therefore evaluate depth under a fixed evidence pack and diagnose Write context only after retrieval is held constant.

4 CoMem

Interface. Let a decoder have L layers and width d, with half-open layer ranges and h_j denoting the input to block j. WRITE divides a document into c-token chunks, runs each through $[0{:}j)$ with positions restarted at zero, and stores $h_j \in \mathbb{R}^{c \times d}$. SELECT returns top-$k$ chunk IDs. READ orders them by document position, assigns fresh contiguous positions to $[\text{BOS}; \{h_j\}; h_j(q)]$, and runs $[j{:}L)$ with full causal cross-pack attention. For j=0, the same path stores token IDs and executes all layers, giving the matched endpoint of the depth axis.

Storage and bounded model work. With head width d_{head}, query-head count n_{q}, and KV-head count n_{kv}, residual-to-full-KV bytes/token at a common dtype are

$$\frac{|h_j|}{|\text{KV}|} = \frac{d}{2Ln_{\text{kv}}d_{\text{head}}} = \frac{n_{\text{q}}}{2Ln_{\text{kv}}}, \tag{1}$$

using $d = n_{\text{q}}d_{\text{head}}$. For Qwen3-8B this is 1/18: 8 KiB rather than 144 KiB/token in bf16. It is about 1 GiB at 128k tokens—far larger than text, so reuse is essential. The model-side Read length is sink + kc + query (nominally 6,657 tokens for k=12, c=512), independent of stored-context length. The persistent store, index, and selector are not bounded.

Overlap-Write. Chunk-local writes omit lower-layer left context. For the diagnostic variant, chunk i is prefixed by its preceding w tokens during Write, but only the current c residuals are stored. Thus persistent bytes, selected-pack length, and per-query Read/decode are unchanged; Write work and edit invalidation grow. The overlap sweep tests how much local context is sufficient to repair independently written states; the default full-suite results retain w=0.

Self-distillation. Following knowledge distillation and low-rank adaptation (Hinton et al., 2015; Hu et al., 2022), a j=12 student matches a frozen j=0 teacher on the teacher's top-64 logit support at every query token. The teacher runs the same chunk-local pack from j=0 with the adapter disabled; the student writes the sink, each context chunk, and the query independently to h_{12}, then resumes the packed query through layers 12–35. If S_t is the teacher's top-64 vocabulary set at query token t, then $p_t = \mathrm{softmax}(z^T_{t,S_t})$ and $q_t = \mathrm{softmax}(z^S_{t,S_t})$: both distributions are renormalized only over the same teacher-selected support, and all logits outside S_t are discarded by this approximate objective. We did not retain the teacher mass captured by S_t. The token-mean loss is

$$\mathcal{L}_{\mathrm{distill}} = 0.6\,\mathrm{KL}(p\|q) + 0.4\,\mathrm{KL}(q\|p), \tag{2}$$

with no cross-entropy term. The rank-32 LoRA (α=64, zero dropout) updates attention and MLP projections in layers 12–35; the Qwen3-8B backbone stays frozen. Training uses 4,000 bf16 DDP steps on 2,048-token PG-19 windows (Rae et al., 2019), without benchmark supervision, retrieval, or synthetic long-context data. Full optimizer, masks, reduction, and artifact details are in the appendix.

Selection and generation. The flagship selector is deterministic token-ID BM25 (Robertson & Zaragoza, 2009), chosen to avoid a separately trained retriever. It uses top-12, hop width four, and three frontier-expansion rounds: round 1 scores the question; later rounds concatenate tokens from newly selected chunks, score unselected chunks, and break ties by document index. Frozen dense retrieval is reported only as a diagnostic. During generation, lower layers cache the query/generation prefix and upper layers cache the selected pack; the retrieved pack is not rerun for each output token. Exact masks, positions, and timing boundaries appear in the appendix.

5 Experiments

Setup. The principal model is frozen Qwen3-8B (Yang et al., 2025a), with j=12, 512-token chunks, iterative BM25 top-12, a BOS sink, and the rank-32 LoRA. Prompting is plain text without chat templates. We evaluate RULER (Hsieh et al., 2024), BABILong (Kuratov et al., 2024), LongEval (Li et al., 2023), six LongBench QA datasets (Bai et al., 2024), and all 1,986 LoCoMo questions (Maharana et al., 2024). Appendices A.3 and B give complete cells, prompts, sample counts, uncertainty, and integrity checks. We keep four timing boundaries separate: selected-pack model Read, store-ready online prefill, Write-inclusive pipeline, and external index/store cost.

5.1 Matched depth reuse

Table 2 is the central comparison. Raw-text replay (j=0) and CoMem receive the same selected chunks in the same order and use the same sink, mask, examples, and LoRA. Skipping the lower 12 layers reduces fixed-pack Read from 931.9 to 664.4 ms (1.403×), but lowers the RULER macro from 99.19 to 96.07. This gives the matched depth frontier: 1.403× faster Read for a 3.12-point quality cost. The continuous-prefix control exactly matches j=0, showing that the upper 24 layers retain sufficient capacity to reproduce replay when supplied compatible states. It therefore turns the observed loss into a Write-interface problem that the following controls can diagnose.

Natural-task transfer is uneven. Relative to matched j=0, j=12 falls sharply on LongEval (97.2 to 69.0), changes modestly on LoCoMo (41.59 to 38.27), and is nearly unchanged

Operating point	Stored object	Online depth	RULER↑	Read	Write	Store/token
Matched raw-text replay (j=0)	token IDs	36 layers	99.19	931.9 ms	per query	∼4–8 B
CoMem + LoRA (j=12)	h_{12}	24 layers	96.07	664.4 ms	once	8,192 B
Continuous-prefix oracle	h_{12}	24 layers	99.19	—	per query	8,192 B

Table 2: **Matched depth reuse delivers** $1.403\times$ **faster Read at a 3.12-point quality cost.** All rows use the same paired 15-cell RULER-B cohort (n=1,500), selected pack, order, sink, mask, examples, and LoRA. The j=0–j=12 gap has paired-bootstrap 95% CI $[2.36, 3.93]$. Read latency uses an approximately 6.5k-token pack from 16k source contexts and three process medians; selection, persistent I/O, reusable Write, and decode are excluded. The continuous-prefix oracle computes lower-layer states jointly with the selected pack and query, exactly recovering 99.19; it therefore shows that the upper 24 layers can reproduce full replay when supplied compatible h_{12} states.

on six LongBench QA datasets (12.31 to 12.15); BABILong varies by task (Table 23). The detailed tables (Tables 22, 24, and 25) include the corresponding measured j=0 reference cells rather than repeating an overview macro in each length or dataset column. They support transfer of the interface, not uniform preservation of task quality. Data-overlap and judge qualifications appear in the appendix and Limitations. The external references also expose important boundaries rather than uniform dominance. In the full-prompt YaRN control, KV-Direct reaches 100 on 128k single-needle and 89 on multikey versus CoMem's 98/93, while CoMem is much stronger on variable tracking (99.0 versus 57.8). At 8k–32k, SnapKV and PyramidKV reach a 99.96 RULER macro after full-prompt prefill; their apparent 72.3 native 15-cell macro is lowered by left-truncated 64k/128k inputs. CoMem's advantage is therefore bounded read without full-prompt prefill and stronger long-range tracking, not universal per-cell accuracy.

5.2 Same-backbone chunk-KV comparison

Table 6 compares CoMem with a minimal faithful CacheBlend-style arm (Yao et al., 2025). The measured dimensions held common are Qwen3-8B, iterative-BM25 top-12 retrieval, 512-token chunks, plain-text prompts, and scorers. Both use the same pack-construction policy, but the retained artifact does not prove identical examples or packs; their length supports also differ, and only CoMem is adapted. The baseline prefills full-depth per-chunk KV, reindexes RoPE when chunks are assembled, and selectively recomputes the highest-deviation context tokens; a real-model r=1 self-test agrees with vanilla full prefill to 2.8×10^{-5} maximum logit difference and 100% top-1 agreement. Increasing r from 0 to .18 improves its 12-cell RULER macro from 67.80 to 74.70, but leaves BABILong qa5 near 49 and LoCoMo substring accuracy near 17. CoMem separately reports 97.05 on 15 RULER cells, 68.67 on seven BABILong qa5 cells, and 23.36 on the full LoCoMo set. Its single residual uses 8 KiB/token rather than 144 KiB/token for full-depth bf16 GQA KV. Thus, as a protocol-aligned diagnostic rather than a matched estimate, full-layer KV still underperforms the residual-plus-upper-recompute path. This is not a native CacheBlend reproduction or an adaptation-matched causal estimate.

5.3 Amortization and equal-latency boundaries

The residual store costs 8,192 bytes/token and must first be written. In the write-once serving harness, CPU-pinned break-even is 8.9–10.9 queries at 32k for generated lengths up to 128 tokens; the full GPU/CPU grid is 5.5–10.9. The 512-token 32k CPU case moves to 94 queries because decode dominates, and the 32k GPU cell has no finite crossover at that output length. At 128k, break-even is 25.8–27.2 queries for GPU placement and 27.6–37.4 for CPU placement through G=128; at 1M tokens it rises to roughly 180–421 queries. Each value follows the displayed measured-time equation and includes fetch, model Read, and decode; for one-off or weakly reused documents, raw text is cheaper.

We then map the boundary at which depth reuse remains attractive. A latency-only calibration freezes maximum budgets of 10 replay chunks and 12 CoMem chunks; selectors may under-fill. With the same iterative-BM25 ranking, raw replay leads by 11.56 points, establishing that the saved depth does not compensate for its representation cost on this tested mixture. Replacing only the replay selector with frozen BGE changes the point difference to -1.00, while the unchanged CoMem arm remains at 53.22. A hierarchical bootstrap that first resamples cells gives $[-18.67, -5.11]$ for BM25 and $[-10.67, 8.33]$ for BGE; leave-one-cell-out ranges are $[-13.13, -9.50]$ and $[-3.50, 1.75]$. The applicable regime is therefore explicit: BM25 replay has a robust advantage, whereas the cross-selector BGE diagnostic is unresolved under task-cell dependence. Depth reuse can buy a larger evidence budget, but whether that budget closes the representation gap depends strongly on retrieval quality. Appendix Table 8 exposes the cohort, weights, generation limits, split, absolute latencies, and bootstrap unit.

5.4 Write-interface diagnosis and repair

Lower-layer Write scope	local Read pos.	document-origin Read pos.
Chunk local	92.5	88.0
Document contextual	100.0	100.0

Table 3: **Bounded context-scope $\times$ Read-position diagnosis.** Paired Qwen3-8B multikey macro over 8k/16k (n=200), with identical retrieval, LoRA, and upper-layer Read. Document context adds 7.5 points with local Read positions and 12.0 with document-origin positions; changing position alone lowers the chunk-local writer by 4.5 points. The $+4.5$ interaction means the factors are not additive. This attribution is limited to the displayed cohort.

Holding retrieval, adapter, and upper-layer Read fixed, the 2×2 diagnostic separates lower-layer attention scope from Read positions. Document-contextual Write closes the gap under either position convention; document-origin positions alone make chunk-local states 4.5 points worse. The $+4.5$ interaction precludes a purely additive explanation and identifies the Write representation—especially missing lower-layer document context—as the dominant tested source of the gap rather than position remapping.

Overlap-Write turns this control into a local deployable intervention: w=32 raises 92.5 to 98.5 and w=128 to 99.0 while preserving stored bytes and per-query work. Its measured wall-clock overhead exceeds the theoretical FLOP ratios, but the result demonstrates that most of the local gap can be repaired with only 32 left-context tokens and no increase in persistent bytes or online Read. A separately trained lower-layer Write adapter reaches the same 98.5 on this cohort; details and paired tests are in Table 15. Both repairs remain synthetic diagnostics rather than natural-task or end-to-end validation.

5.5 Efficiency and timing boundaries

The main store-ready reference is the H20 Pareto harness: stock Qwen3-8B without the CoMem adapter prefills the full 128k source in 50.59 s, whereas adapted CoMem processes its selected pack in 1.32 s (Table 4). This composed operating point combines bounded selection with depth reuse and excludes one-time document Write and external-store fetch; because its dense arm is LoRA-off, it is not a same-adapter comparison. Finally, the same-adapter L20A pipeline comparison processes the full source in 6.035 s and CoMem in 2.202 s ($2.74\times$), with the latter including document Write, CPU BM25 retrieval, query Write, and upper-layer Read but excluding BM25-index construction and external I/O (Table 30). The two measurements therefore separate store-ready online work from a Write-inclusive same-adapter pipeline rather than forming interchangeable speedup denominators; the appendix retains the separate H20 cohort for completeness. The Write-inclusive comparison is length-dependent: CoMem is slower at 8k and 16k (0.53/0.81×) and crosses dense prefill only from 32k onward, reaching 2.74× at 128k.

On the fixed-pack matched harness, the 267.5 ms Read saving becomes only 1.07–1.09× end-to-end once 2.76–2.86 s autoregressive decode is included. This is the expected decode bottleneck for bounded-read systems; speculative decoding and prefix caching are orthogonal and not measured here.

6 Conclusion

CoMem exposes prepaid depth as a reusable-context axis. Prepaying 12 Qwen3-8B layers gives a 1.403× Read speedup at a 3.12-point RULER cost; matched replay and Write controls delimit the trade-off.

Limitations

CoMem is evaluated mainly on one Qwen3-8B model, a lexical selector, and English benchmarks. Cross-scale and sparse-MoE results test implementation portability, not matched replication, and some large-model cells are smaller. The flagship adapter is one batch-8 run; two matched-data runs use effective batch 3, so they conflate seed and optimization noise. We still lack same-backbone, same-hardware implementations of the broader PIC and learned modular-cache families. The included CacheBlend-style experiment is a minimal faithful full-depth chunk-KV control with global RoPE reindexing and selective recomputation, not the complete native CacheBlend scheduler, cache manager, or production serving stack; it is training-free whereas CoMem uses a distilled adapter. Published systems still differ in model, task, training, storage tier, and timing boundary, so Table 1 remains structural rather than a cross-paper ranking. The controlled depth evidence is the $j=0 \rightarrow 12$ endpoint; separately trained splits form a deployment curve, not a compute-matched causal frontier.

The workload assumes repeated queries over a stable corpus. Residual storage is linear and much larger than text, selector/store latency grows with corpus size, and model-specific states generally require rewriting after changes to the backbone, tokenizer, split, adapter, or lower-layer implementation. One-off, low-reuse, or frequently edited documents may favor raw text; Overlap-Write can also invalidate a neighboring chunk after an edit. Latency and break-even depend on hardware, kernels, batch size, storage placement, and timing boundaries. The large-output crossover can be unstable when decode leaves only a small per-query latency difference; serving reports single-query medians rather than concurrent throughput or p95/p99 tails. Quantization, eviction, update contention, and production multi-tenancy are not evaluated. The 50.59/1.32 s and 6.035/2.202 s efficiency figures come from separate H20 and L20A harnesses with different adapter and inclusion boundaries; they are labeled separately and must not be combined into one speedup.

Quality remains task-, selector-, and evidence-budget-dependent. Top-k limits coverage; lexical retrieval transfers poorly to low-overlap and non-whitespace settings; and residual readout can fail after a retrieval hit. Overlap-Write is validated only on paired synthetic multikey instances, without a natural-task or Write-inclusive repaired frontier. The trained lower-layer Write adapter is evaluated on the same synthetic cohort and therefore remains a mechanism upper-bound rather than evidence of natural-task repair. Beyond-native-window results are positional stress tests. In the equal-latency mixture, BM25 replay wins robustly, while the frozen-BGE difference spans both signs; cells are equally weighted and the 100 LoCoMo items come from one conversation, preventing conversation-cluster inference. Thus this diagnostic does not establish a selector-independent advantage.

LoCoMo uses a mutable, undated `gpt-4o` judge; saved decisions and a 200-item DeepSeek-V3 audit preserve ordering but not exact reproducibility under a fixed snapshot. PG-19 overlaps InfiniteBench long-book support, so that comparison was removed. A later audit localizes LongBench overlap to NarrativeQA and finds at most a 0.13-point macro-F1 change after strict clean-subset recomputation; LoCoMo is clean and LongEval is synthetic by construction. This supports robustness to detected near-verbatim overlap, not a guarantee against semantic paraphrase or exposure in other pretraining corpora. Finally, probes measure readout compatibility rather than the location or causal necessity of "understanding" or factual knowledge. Long-form, code, multilingual, multimodal, and dynamic-memory use remain open.

Ethical Considerations

CoMem inherits the underlying model's hallucination, bias, unsafe-generation, and misuse risks; cheaper access to long histories can scale both beneficial and harmful applications. Retrieved or cached material may disclose sensitive or unauthorized text, and residual tensors may permit inversion or membership inference, although we do not test these attacks. Deployments should protect the store at least as strongly as source text through authorization, tenant isolation, encryption, auditing, pre-Write filtering/redaction, output monitoring, and verifiable deletion; large persistent stores and Overlap-Write also require careful update invalidation. Bounded reads reduce per-query compute, but training and evaluation still consume energy; measured final-run hardware and compute appear in Appendix A.5.

We collect no new human-subject data or annotations. Experiments use released models, corpora, and benchmarks; LoCoMo conversations are model-generated. The anonymous artifact excludes model weights, benchmark text, credentials, private API metadata, and raw private responses. Subject to upstream terms, it provides the adapter, evaluation code, permissible prediction artifacts or hashes, judge prompts/parsed decisions, timing records, licenses, and provenance needed to audit the reported results.

References

Shubham Agarwal, Sai Sundaresan, Subrata Mitra, Debabrata Mahapatra, Archit Gupta, Rounak Sharma, Nirmal Joshua Kapu, Tong Yu, and Shiv Saini. Cache-Craft: Managing chunk-caches for efficient retrieval-augmented generation. *Proceedings of the ACM on Management of Data*, 2025. doi: 10.1145/3725273.

Yushi Bai, Xin Lv, Jiajie Zhang, Hongchang Lyu, Jiankai Tang, Zhidian Huang, Zhengxiao Du, Xiao Liu, Aohan Zeng, Lei Hou, Yuxiao Dong, Jie Tang, and Juanzi Li. LongBench: A bilingual, multitask benchmark for long context understanding. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 3119–3137, 2024. doi: 10.18653/v1/2024.acl-long.172.

Harsh Vardhan Bansal. LLMCache: Layer-wise caching strategies for accelerated reuse in transformer inference. *arXiv preprint arXiv:2512.16843*, 2025.

Zefan Cai, Yichi Zhang, Bofei Gao, Yuliang Liu, Tianyu Liu, Keming Lu, Wayne Xiong, Yue Dong, Baobao Chang, Junjie Hu, and Wen Xiao. PyramidKV: Dynamic KV cache compression based on pyramidal information funneling. In *Conference on Language Modeling*, 2025.

Chuangtao Chen, Grace Li Zhang, Xunzhao Yin, Cheng Zhuo, Bing Li, and Ulf Schlichtmann. KV Packet: Recomputation-free context-independent KV caching for LLMs. *arXiv preprint arXiv:2604.13226*, 2026.

Sabri Eyuboglu, Ryan Ehrlich, Simran Arora, Neel Guha, Dylan Zinsley, Emily Liu, Atri Rudra, James Y. Zou, Azalia Mirhoseini, and Christopher Ré. Cartridges: Lightweight and general-purpose long context representations via self-study. In *International Conference on Learning Representations*, volume 2026, pp. 42642–42687, 2026.

Shiwei Gao, Youmin Chen, and Jiwu Shu. Fast state restoration in LLM serving with HCache. In *Proceedings of the Twentieth European Conference on Computer Systems (EuroSys)*, pp. 128–143, 2025. doi: 10.1145/3689031.3696072.

In Gim, Guojun Chen, Seung-seob Lee, Nikhil Sarda, Anurag Khandelwal, and Lin Zhong. Prompt cache: Modular attention reuse for low-latency inference. In *Proceedings of Machine Learning and Systems*, volume 6, pp. 325–338, 2024.

Aaron Grattafiori et al. The Llama 3 herd of models. *arXiv preprint arXiv:2407.21783*, 2024.

Momchil Hardalov, Gonzalo Iglesias, and Adrià de Gispert. Cartridges at scale: Training modular KV caches over large document collections. *arXiv preprint arXiv:2606.04557*, 2026.

Geoffrey Hinton, Oriol Vinyals, and Jeff Dean. Distilling the knowledge in a neural network. *arXiv preprint arXiv:1503.02531*, 2015.

Cheng-Ping Hsieh, Simeng Sun, Samuel Kriman, Shantanu Acharya, Dima Rekesh, Fei Jia, Yang Zhang, and Boris Ginsburg. RULER: What's the real context size of your long-context language models? In *First Conference on Language Modeling*, 2024.

Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations*, 2022.

Junhao Hu, Wenrui Huang, Weidong Wang, Haoyi Wang, Tiancheng Hu, Qin Zhang, Hao Feng, Xusheng Chen, Yizhou Shan, and Tao Xie. EPIC: Efficient position-independent caching for serving large language models. In *International Conference on Machine Learning*, 2025.

Chao Jin, Zili Zhang, Xuanlin Jiang, Fangyue Liu, Shufan Liu, Xuanzhe Liu, and Xin Jin. RAGCache: Efficient knowledge caching for retrieval-augmented generation. *ACM Transactions on Computer Systems*, 44:1–27, 2025. doi: 10.1145/3768628.

Yuri Kuratov, Aydar Bulatov, Petr Anokhin, Ivan Rodkin, Dmitry Sorokin, Artyom Sorokin, and Mikhail Burtsev. BABILong: Testing the limits of LLMs with long context reasoning-in-a-haystack. In *Advances in Neural Information Processing Systems*, volume 37, pp. 106519–106554, 2024. doi: 10.52202/079017-3381.

Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Kuettler, Mike Lewis, Wen-tau Yih, Tim Rocktaschel, Sebastian Riedel, and Douwe Kiela. Retrieval-augmented generation for knowledge-intensive NLP tasks. In *Advances in Neural Information Processing Systems*, volume 33, 2020.

Dacheng Li, Rulin Shao, Anze Xie, Ying Sheng, Lianmin Zheng, Joseph E. Gonzalez, Ion Stoica, Xuezhe Ma, and Hao Zhang. How long can open-source LLMs truly promise on context length? LMSYS Org Blog, 2023. URL https://lmsys.org/blog/2023-06-29-longchat.

Yuhong Li, Yingbing Huang, Bowen Yang, Bharat Venkitesh, Acyr Locatelli, Hanchen Ye, Tianle Cai, Patrick Lewis, and Deming Chen. SnapKV: LLM knows what you are looking for before generation. In *Advances in Neural Information Processing Systems*, volume 37, pp. 22947–22970, 2024. doi: 10.52202/079017-0722.

Manlai Liang, Mandi Liu, Jiangzhou Ji, Huaijun Li, Haobo Yang, Yaohan He, and Jinlong Li. ILRe: Intermediate layer retrieval for context compression in causal language models. *arXiv preprint arXiv:2508.17892*, 2025.

Shih-Ting Lin, Ashish Sabharwal, and Tushar Khot. Readonce transformers: Reusable representations of text for transformers. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics*, 2021. doi: 10.18653/v1/2021.acl-long.554.

Akide Liu, Jing Liu, Zizheng Pan, Yefei He, Gholamreza Haffari, and Bohan Zhuang. MiniCache: KV cache compression in depth dimension for large language models. In *Advances in Neural Information Processing Systems*, volume 37, pp. 139997–140031, 2024. doi: 10.52202/079017-4443.

Songshuo Lu, Hua Wang, Yutian Rong, Zhi Chen, and Yaohua Tang. TurboRAG: Accelerating retrieval-augmented generation with precomputed KV caches for chunked text. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pp. 6588–6601, 2025. doi: 10.18653/v1/2025.emnlp-main.334.

Dongyang Ma, Yan Wang, and Tian Lan. Block-attention for efficient prefilling. In *International Conference on Learning Representations*, 2025.

Adyasha Maharana, Dong-Ho Lee, Sergey Tulyakov, Mohit Bansal, Francesco Barbieri, and Yuwei Fang. Evaluating very long-term conversational memory of LLM agents. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 13851–13870, 2024. doi: 10.18653/v1/2024.acl-long.747.

João Monteiro, Étienne Marcotte, Pierre-André Noël, Valentina Zantedeschi, David Vázquez, Nicolas Chapados, Christopher Pal, and Perouz Taslakian. XC-Cache: Cross-attending to cached context for efficient LLM inference. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, 2024. doi: 10.18653/v1/2024.findings-emnlp.896.

Kaleem Ullah Qasim, Jiashu Zhang, Muhammad Kafeel Shaheen, Razan Alharith, and Heying Zhang. The residual stream is all you need: On the redundancy of the KV cache in transformer inference. *arXiv preprint arXiv:2603.19664*, 2026.

Jack W. Rae, Anna Potapenko, Siddhant M. Jayakumar, and Timothy P. Lillicrap. Compressive transformers for long-range sequence modelling. *arXiv preprint arXiv:1911.05507*, 2019.

Stephen Robertson and Hugo Zaragoza. The probabilistic relevance framework: BM25 and beyond. *Foundations and Trends in Information Retrieval*, 4(1–2):1–174, 2009. doi: 10.1561/1500000019.

Jon Saad-Falcon, Amanpreet Singh, Luca Soldaini, Mike D'Arcy, Arman Cohan, and Doug Downey. Embedding recycling for language models. In *Findings of the Association for Computational Linguistics: EACL 2023*, 2023. doi: 10.18653/v1/2023.findings-eacl.145.

Zhenmei Shi, Yifei Ming, Xuan-Phi Nguyen, Yingyu Liang, and Shafiq Joty. Discovering the gems in early layers: Accelerating long-context LLMs with 1000x input token reduction. In *Findings of the Association for Computational Linguistics: ACL 2026*, pp. 13839–13857, 2026. doi: 10.18653/v1/2026.findings-acl.677.

Woomin Song, Sai Muralidhar Jayanthi, Srikanth Ronanki, Kanthashree Mysore Sathyendra, Jinwoo Shin, Aram Galstyan, Shubham Katiyar, and Sravan Babu Bodapati. Compress, gather, and recompute: REFORMing long-context processing in transformers. *arXiv preprint arXiv:2506.01215*, 2025.

Sijun Tan, Xiuyu Li, Shishir Patil, Ziyang Wu, Tianjun Zhang, Kurt Keutzer, Joseph E. Gonzalez, and Raluca Ada Popa. LLoCO: Learning long contexts offline. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pp. 17605–17621, 2024. doi: 10.18653/v1/2024.emnlp-main.975.

Tencent Hunyuan Team. Hy3: An open 295b mixture-of-experts model with 21b activated parameters. https://huggingface.co/tencent/Hy3, 2026.

Jiahao Wang, Weiyu Xie, Mingxing Zhang, Boxing Zhang, Jianwei Dong, Yuening Zhu, Chen Lin, Jinqi Tang, Yaochen Han, Zhiyuan Ai, Xianglin Chen, Yongwei Wu, and Congfeng Jiang. From prefix cache to fusion RAG cache: Accelerating LLM inference in retrieval-augmented generation. *arXiv preprint arXiv:2601.12904*, 2026.

Qian Wang, Zahra Yousefijamarani, Morgan Lindsay Heisler, Rongzhi Gu, Bai Xiaolong, Shan Yizhou, Wei Zhang, Wang Lan, Ying Xiong, Yong Zhang, and Zhenan Fan. MEPIC: Memory efficient position independent caching for LLM serving. *arXiv preprint arXiv:2512.16822*, 2025.

Weizhi Wang, Li Dong, Hao Cheng, Xiaodong Liu, Xifeng Yan, Jianfeng Gao, and Furu Wei. Augmenting language models with long-term memory. In *Advances in Neural Information Processing Systems*, 2023.

Yu Wang, Yifan Gao, Xiusi Chen, Haoming Jiang, Shiyang Li, Jingfeng Yang, Qingyu Yin, Zheng Li, Xian Li, Bing Yin, Jingbo Shang, and Julian McAuley. MemoryLLM: Towards self-updatable large language models. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pp. 50453–50466. PMLR, 2024.

Chaojun Xiao, Pengle Zhang, Xu Han, Guangxuan Xiao, Yankai Lin, Zhengyan Zhang, Zhiyuan Liu, and Maosong Sun. InfLLM: Training-free long-context extrapolation for LLMs with an efficient context memory. In *Advances in Neural Information Processing Systems*, volume 37, pp. 119638–119661, 2024a. doi: 10.52202/079017-3801.

Guangxuan Xiao, Yuandong Tian, Beidi Chen, Song Han, and Mike Lewis. Efficient streaming language models with attention sinks. In *International Conference on Learning Representations*, 2024b.

Hui Xie, Peng Xiao, Yutong Deng, Shuoran Dou, Jian Yang, and Jinyang Guo. SemPIC: Learning semantic position-independent KV caches. *arXiv preprint arXiv:2607.28069*, 2026.

Peng Xu, Wei Ping, Xianchao Wu, Lawrence McAfee, Chen Zhu, Zihan Liu, Sandeep Subramanian, Evelina Bakhturina, Mohammad Shoeybi, and Bryan Catanzaro. Retrieval meets long context large language models. In *International Conference on Learning Representations*, 2024.

An Yang et al. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*, 2025a.

Xinyu Yang, Tianqi Chen, and Beidi Chen. APE: Faster and longer context-augmented generation via adaptive parallel encoding. *arXiv preprint arXiv:2502.05431*, 2025b.

Jiayi Yao, Hanchen Li, Yuhan Liu, Siddhant Ray, Yihua Cheng, Qizheng Zhang, Kuntai Du, Shan Lu, and Junchen Jiang. CacheBlend: Fast large language model serving for RAG with cached knowledge fusion. In *Proceedings of the Twentieth European Conference on Computer Systems*, pp. 94–109, 2025. doi: 10.1145/3689031.3696098.

Zhenyu Zhang et al. H2O: Heavy-hitter oracle for efficient generative inference of large language models. In *NeurIPS*, 2023.

A Additional Results and Ablations

This appendix contains the detailed task tables, mechanism ablations, and cross-scale generality results summarized in the main paper. Appendix B separately documents aggregation, uncertainty, and the LoCoMo judge protocol.

A.1 Systems accounting and equal-latency audit

Detailed interface positioning. Text RAG is the matched $j=0$ endpoint: it stores token IDs and reruns every layer after selection. Reusable-representation methods fix a compressed or intermediate interface and adapt the downstream computation. Semantic caches match similar inputs against per-layer activation banks, whereas hidden-state restoration reconstructs standard layer-wise KV. Chunk-KV and PIC systems store full-depth KV objects and vary linking or repair; learned modular KV methods move more document computation into training or compilation. CoMem instead stores one document residual at a chosen split and directly executes the native suffix. These families differ in backbone, training, storage tier, and timing boundary, so Table 1 is structural rather than a numerical ranking; Table 6 provides the focused same-backbone diagnostic.

Secondary dense-context operating point. When the persistent store already exists, the select-first path reads a fixed 6,657-token pack at every source length. Table 4 reports 1.32 s and 18.7 GB at 128k versus 50.59 s and 50.0 GB for stock dense prefill on the same H20. The resulting 38.3× combines bounded selection with depth reuse; it is neither the incremental depth effect nor an end-to-end repeated-query comparison.

Length	Dense prefill	CoMem prefill	Speedup	Dense peak	CoMem peak
8k	1.10 s	0.94 s	1.17×	18.8 GB	18.6 GB
32k	5.94 s	1.09 s	5.45×	25.0 GB	18.7 GB
128k	50.59 s	1.32 s	38.3×	50.0 GB	18.7 GB

Table 4: **Store-ready online-prefill operating points.** Measurements use one H20, Qwen3-8B, bf16/SDPA, and the median of three runs after one warmup. Dense is the stock, LoRA-off Qwen3 path and processes the complete source; CoMem reads a fixed 6,657-token top-12 pack with its rank-32 LoRA. The persistent store is already built, so these values isolate the store-ready online path and exclude one-time Write and external-store fetch. The matched incremental depth effect is the separate 1.403× result in Table 2; the Write-inclusive same-adapter L20A cohort appears in Table 30. The 50.59 s dense row is therefore a stock-dense master reference, not a same-adapter control.

Equal-latency audit. Table 8 reconstructs the complete saved protocol behind Table 7, including the nine cells, equal cell weights, disjoint latency calibration, generation limits, absolute TTFT, hardware, storage assumptions, all cell deltas, dependence-aware bootstraps, leave-one-cell-out sensitivity, and the original pooled-IID interval.

Content-depth probe protocol. For SST2, WiC, and RTE we extract every layer once from a fixed stratified pool (up to 3,000 train examples), then fit L2 logistic regressions on five independent 60/20/20 train/dev/test splits. Features are standardized; $C \in \{0.1, 1, 10\}$ is selected on development data. For layer accuracy $a(l)$, the knee is $\min\{l : a(l) \geq 0.98 \max_l a(l)\}$. The native readout instead applies the model's own final norm and LM head to each layer using fixed verbalizers. Lexical uni/bi-grams, token length, random labels, majority class, and balanced training serve as controls. On SST2, probe peak is about 0.90 versus 0.71 lexical, 0.56 position/majority, and 0.54 random-label accuracy; WiC/RTE are lower-selectivity and have wider per-task intervals. The reported confidence intervals are Student-t intervals over task–split knees, not training-seed uncertainty.

Store	G=1	G=32	G=128	G=512
32k, CPU-pinned	8.9	9.2	10.9	94.0
32k, GPU-resident	8.4	7.7	5.5	∞
128k, CPU-pinned	27.6	29.7	37.4	520.1
128k, GPU-resident	25.8	26.8	27.2	164.2
1M, CPU-pinned	198.2	266.7	421.3	302.6
1M, GPU-resident	180.2	183.0	190.6	574.7

Table 5: **Measured repeated-query break-even $Q^\star$ over the full serving grid.** G is generated tokens; a cell is the number of queries after which cumulative CoMem time is lower than matched j=0 raw-text replay. Both arms share selection, example, and pack; selection cancels in $Q^\star = (W_{\text{CoMem}} - I_{j=0})/(T_{j=0} - T_{\text{CoMem}})$. The 32k CPU cell uses W_{CoMem}=2.253 s and a 2.16 ms raw-token index build. Values derive from three process medians and include store fetch, model Read, and decode. The 18 per-process records and aggregate are released with SHA-256 manifests. ∞ means CoMem's measured per-query time is not lower, so no finite query count amortizes its one-time Write in that cell. Large-G values are sensitive because decode leaves a small per-query denominator; they are workload-specific measurements rather than a monotonic scaling law.

Method / recompute	RULER	BABI-qa5	LoCoMo	Store/tok.
CacheBlend-style, r=0	67.80	49.33	17.07	144 KiB
CacheBlend-style, r=.10	72.57	48.50	16.77	144 KiB
CacheBlend-style, r=.15	73.78	49.83	17.07	144 KiB
CacheBlend-style, r=.18	74.70	49.17	17.37	144 KiB
CoMem (j=12)	97.05	68.67	23.36	8 KiB

Table 6: **Same-backbone full-depth chunk-KV comparison.** The baseline uses independently prefetched 36-layer KV, global RoPE reindexing, and selective recomputation; its r=1 self-test matches full prefill. Both arms use Qwen3-8B, 512-token chunks, iterative-BM25 top-12, plain text, and the same scorers. However, CacheBlend reports 12 RULER cells at 4k–32k and six BABI-qa5 cells at 0k–16k, whereas the CoMem entries summarize 15 cells at 8k–128k and seven cells at 0k–32k. LoCoMo has 1,986 questions for both, but the artifact does not prove itemwise pack identity. The training-free baseline and adapted CoMem are therefore protocol-aligned diagnostics, not paired or adaptation-matched estimates. Full-depth bf16 GQA KV costs 147,456 bytes/token versus 8,192 for CoMem. The recorded BABILong `resume_j=6` is a host default; CacheBlend executes all 36 layers.

Replay selector	$k_{\max}$	Replay / CoMem	Δ	Hier. 95% CI	LOCO range
Iterative BM25	10	64.78 / 53.22	−11.56	[−18.67, −5.11]	[−13.13, −9.50]
Frozen BGE	10	54.22 / 53.22	−1.00	[−10.67, 8.33]	[−3.50, 1.75]

Table 7: **Equal-latency quality exposes selector choice as a first-order frontier variable.** A latency-only calibration fixes replay budget 10 against CoMem budget 12 within a predeclared ±5% band. Quality is the equal-weight mean of nine 100-example task–length cells; Δ is CoMem minus replay. The BM25 replay advantage is robust to hierarchical and leave-one-cell-out analyses. The frozen-BGE point estimate favors replay by one point, but both dependence-aware ranges span zero. Appendix Table 8 gives the complete protocol and sensitivity analyses.

Protocol item	Setting	Auditable detail
Quality cohort	9 cells, n=100/cell	BABILong qa1/qa2 at 4k and 16k; LongEval at 8k and 16k; RULER `niah_multikey_1` at 8k and 16k; first 100 LoCoMo items. Each cell contributes one equally weighted accuracy mean. The retained LoCoMo slice is entirely conversation 0, so its 100 questions cannot support a conversation-cluster bootstrap.
Generation/scoring	Greedy, no chat	BABILong: 20 tokens and official `compare_answers`; LongEval: 16 tokens and numeric exact match; RULER: 48 tokens and official `string_match_all`; LoCoMo diagnostic: 48 tokens and deterministic local substring/F1-threshold or abstention accuracy, not the GPT-4o headline judge.
Calibration split	Latency only, disjoint	Three RULER multikey 32k documents at reserved indices 900–902 (`PYTHONHASHSEED=0`, experiment seed 42), disjoint from quality indices 0–99. Integer $k \in \{2,4,6,8,10,12,14,16,20,24\}$ is selected only by TTFT within $\pm$5% of the CoMem budget-12 anchor. Here k is a maximum chunk budget; iterative BM25 may under-fill.
Timing	One NVIDIA H20; bf16/SDPA	Three independent processes per k, five warmups and 20 repetitions each; reported TTFT is the median of process medians. It includes online selection, raw/residual fetch, sink/query Write, and model prefill to first logits; decode, one-time document Write, and index construction are excluded. BM25 replay budget 10 is 703.7 ms versus CoMem 698.3 ms (GPU) / 699.8 ms (CPU); BGE replay is 697.8 ms versus 700.5 / 702.2 ms.
Cell deltas: BM25	CoMem(12) minus replay(10), points	qa1: -6 (4k), -17 (16k); qa2: -3, -10; LongEval: -26 (8k), -28 (16k); RULER: -6, -9; LoCoMo: $+1$. Aggregate -11.56.
Cell deltas: BGE	CoMem(12) minus replay(10), points	qa1: -5 (4k), -21 (16k); qa2: 0, -23; LongEval: 0 (8k), $+16$ (16k); RULER: $+1$, $+19$; LoCoMo: $+4$. Aggregate -1.00.
Dependence-aware inference	100,000 paired bootstrap replicates	**Stratified fixed-cell:** keep all nine cells and resample exactly n=100 pairs inside each cell; intervals are $[-14.33, -8.78]$ (BM25) and $[-4.56, 2.56]$ (BGE). **Hierarchical:** resample nine cell labels, then resample n=100 pairs within every selected occurrence; intervals are $[-18.67, -5.11]$ and $[-10.67, 8.33]$. **Leave-one-cell-out:** $[-13.13, -9.50]$ (all 9 negative) and $[-3.50, 1.75]$ (7 negative, 2 positive). Deterministic base seed is 20260804.
IID sensitivity	Original v5 analysis	Pool all 900 paired differences and resample 900 pairs: $[-14.44, -8.67]$ (BM25) and $[-4.67, 2.67]$ (BGE). These intervals are retained for continuity but are not the task-mixture inference.

Table 8: **Protocol-complete equal-latency audit and dependence-aware reanalysis.** Phase B intentionally uses frozen BGE for raw replay and iterative BM25 for CoMem, so it tests selector sensitivity rather than a same-pack depth effect. The reanalysis reads only saved per-example scores; it runs no model. The anonymous release contains score-only exports (no benchmark text, gold answers, or predictions), the script, result JSON, and SHA-256 manifests.

A.2 Depth, selection, and interface ablations

Split depth and self-distillation. The probe values plotted in Figure 2 use five stratified data splits for Qwen and OLMo plus a matched-support Llama SST2 result; the protocol is specified above. Table 9 gives the frozen $j = 0/6/9/12$ quality sweep and same-j distilled control; Table 12 gives separately distilled $j = 6/9/12/18$ deployment points. Table 10 provides a separate same-j RULER control. Finally, Table 11 uses an HCache-style checkpoint path only as an interface-adaptation diagnostic, showing that the readout repair transfers beyond CoMem.

Configuration	Split j	LoRA	LoCoMo Judge	BABILong mean	
				qa1 / qa2	qa5
Matched raw-text replay	0	no	**41.59**	66.3 / 38.9	63.3
CoMem frozen	6	no	32.78	44.6 / 22.6	53.9
CoMem frozen	9	no	29.15	42.4 / 19.6	55.6
CoMem frozen	12	no	24.52	33.4 / 18.0	60.3
CoMem distilled	12	yes	38.27	55.6 / 27.0	**68.7**

Table 9: Controlled comparison of retrieval, split depth, and self-distillation using Qwen3-8B. All configurations follow the same plain-text prompting protocol, with model-specific chat templating disabled (`chat_template=False`), and retrieve the same top-12 pack. For the frozen configurations, readout fidelity decreases as the split moves deeper. At the same split j=12, the rank-32 LoRA improves the LoCoMo Judge score by 13.75 points and the BABILong qa1, qa2, and qa5 means by 22.2, 9.0, and 8.4 points, respectively. BABILong entries are rounded task means from the underlying 21 task–length cells; the raw-cell j=0 macro is 56.14. The j=0 configuration performs full-depth recomputation over the retrieved pack, providing a selected-pack reference without lower-layer reuse and thus no depth-based reduction in query-time computation.

Task	Adapter	8k	16k	32k
niah_single_2	off	36	7	22
	on	**100**	**100**	**100**
niah_multikey_1	off	8	3	1
	on	**91**	**95**	**92**

Table 10: Same-j=12 RULER adapter ablation (n=100/cell). Each on/off pair shares task, length, selector, and cohort.

Path	LoRA	LoCoMo
HCache-style (j=12)	no	13.29
HCache-style (j=12)	yes	**31.17**

Table 11: Retrieval-free HCache-style checkpoint control. The shared split and harness isolate interface adaptation, showing that the readout repair transfers beyond CoMem.

Selection. Table 13 compares oracle, BM25, recency, and reader attention. BM25 is near oracle on the lexical needle tasks and is more reliable than the tested training-free alternatives at long lengths. Variable tracking instead requires iterative expansion along literal variable links; a three-round CPU microbenchmark costs 188.8 ms at 128k, versus 41.0 ms for one-shot BM25, without an additional model forward. This is a low-overhead selector in the tested regime, not a constant-time one: lookup grows approximately linearly with store size (Table 26). A frozen BGE-large-en-v1.5 retriever is also evaluated on a focused cohort: it preserves strong hit-conditional reader accuracy but loses recall as context grows. We

Quality

Split j	LoRA params	RULER B	Full − deployed (95% CI)	LoCoMo
6	72.745M	98.29	0.91 [0.49, 1.39]	40.38
9	65.470M	97.55	1.65 [1.04, 2.31]	39.02
12	58.196M	96.07	3.12 [2.36, 3.93]	38.27
18	43.647M	55.41	43.79 [41.77, 45.85]	28.65

Systems

Split j	Read (ms)	Read speedup	Write (ms)
6	830.3	1.166×	133.4
9	748.3	1.273×	196.9
12	664.4	1.403×	—
18	499.5	1.807×	390.3

Table 12: Multi-depth deployment curve. Each split has a separately trained rank-32 adapter on layers $[j{:}36)$ under the same data order, seed, steps, optimizer, and KL recipe. RULER uses Cohort B (n=1,500 paired examples); gap CIs are paired bootstrap intervals against full replay with that depth's adapter, and all McNemar tests reject equality ($p \leq 3.05 \times 10^{-5}$). LoCoMo reports the full-set GPT-4o judge point estimate (n=1,986). Read/Write use a fixed 16k same-pack input and three process medians; retrieval, I/O, and decode are excluded, and total latency remains decode-dominated. The matched fixed-pack Write value was not retained for the earlier j=12 artifact. Adapter spans and parameter counts differ, so this is not a training-compute-matched depth ablation.

do not train or tune a retriever, so our results establish lexical and frozen-dense operating points rather than selector optimality.

Task	Len	BM25	Recency	ReaderAttn	Oracle
niah_single	8k	100	100	100	100
	16k	100	100	100	100
	32k	100	82	73	100
niah_multikey	8k	99	98	97	100
	16k	99	88	90	100
	32k	99	54	60	100
var-track	8k	99.4	99.2	99.8	N/A
	16k	92.6	92.4	92.4	N/A
	32k	32.0	41.2	27.8	N/A

Table 13: Single-pass RULER selector sweep (n=100; no chat). Entries are the peak over $k \in \{4, 8, 12, 16, 24\}$. Oracle confirms that the needle-task gap can be selection-limited; it is undefined for variable tracking.

Measure	qa1	qa2	LE	MK	LoCoMo
Dense recall@12	.52	.45	.58	.81	.70
Raw reader quality	.45	.31	.57	.82	.18

Table 14: Frozen BGE-large-en-v1.5 diagnostic at 16k (LoCoMo at native length), with the same chunks and a Qwen3-8B j=0 reader (n=100/cell). Long-context recall can be the bottleneck even when hit-conditional reading is strong. LE/MK denote LongEval/RULER multikey.

Write-context factorization. The complete 2×2 control is reported in the main paper (Table 3). It confirms that the position-only intervention does not repair chunk-local states and that context and position interact. The full overlap and trained-Write sweep is reported in Table 15.

Writer	Context	Macro↑	Write FLOPs
Chunk local (w=0)	current chunk	92.5	1.000×
Overlap (w=32)	32-token left prefix	98.5	1.057×
Overlap (w=64)	64-token left prefix	98.5	1.115×
Overlap (w=128)	128-token left prefix	**99.0**	1.229×
Trained chunk-local Write†	current chunk	98.5	1.000×
Document-context oracle	full prefix	100.0	$O(N)$ forward

Table 15: **A 32-token Write overlap recovers 6.0 of the 7.5-point local context gap.** Paired Qwen3-8B results on RULER `niah_multikey_1` at 8k and 16k (n=200 pooled). For w=32, the gain over chunk-local Write has 95% CI $[3.0, 9.5]$. Every deployable overlap row preserves persistent storage and per-query Read/decode; only one-time Write work and edit invalidation increase. FLOPs are theoretical lower-12 Write ratios, while measured wall-clock overhead is larger because execution remains chunk-level. †A separately trained lower-layer Write LoRA (layers 0–11) reaches 98.5 at step 2000; its gain over chunk-local Write is significant (p=5×10^{-4}, McNemar), while its residual gap to the document-context oracle is not (p=0.25). This uses the same paired synthetic cohort and is an upper-bound diagnostic, not a natural-task validation.

Read interaction and chunk size. Table 17 compares full cross-chunk attention with block-diagonal reuse. Table 18 reports the chunk-size sweep; its entries are point estimates rather than a multi-seed stability ranking.

Method	RULER task	8k	16k	32k	64k	128k
CoMem	niah_single_2	100.0	100.0	99.0	99.0	100.0
	var-track	96.6	97.6	98.8	99.0	95.8
StreamingLLM	niah_single	85.0	36.0	20.0	10.0	2.0
	var-track	41.8	2.2	0.6	0.2	0.8

Table 16: Equal-budget, no-chat RULER comparison (n=100/cell). CoMem and StreamingLLM use the same nominal budget and scorer but not paired examples; CoMem retrieves about 6.5k relevant tokens, whereas StreamingLLM keeps recent tokens.

Task	Full attn.	Block-diag.	Δ
RULER niah_single	100 / 100	100 / 96	0 / +4
RULER niah_multikey	**96 / 94**	60 / 32	+36 / +62
BABILong qa2	36 / 20	17 / 13	+19 / +7
BABILong qa5	78 / 69	78 / 55	0 / +14

Table 17: Full cross-chunk attention versus block-diagonal reuse (8k/16k, no chat). RULER uses n=50 and BABILong n=100 with identical top-12 retrieval. Joint recomputation matters most for multi-fact tasks.

chunk	read tokens	uncached step (s/token)	8k	16k	32k	64k
128	1,665	0.19	91.0	90.0	81.0	85.0
256	3,329	0.37	80.0	90.0	90.0	**94.0**
512	6,657	0.72	89.0	**95.0**	**97.0**	**94.0**
1024	13,313	1.60	**100.0**	89.0	92.0	**94.0**

Table 18: Chunk-size trade-off on RULER `niah_multikey` (n=100, no chat). Packed-step cost grows approximately linearly with read length; accuracy values are point estimates, not a multi-seed claim.

Method	RULER↑ 15-cell	LongEval↑ 8k–128k	LongBench↑ 6-QA F1	BABILong↑ qa1/2/5 macro	LoCoMo Judge↑ full set
CoMem + LoRA (j=12)	97.05	69.0	12.15	50.43	38.27
Unmatched descriptive references; do not compare columns as a ranking					
KV-Direct	78.80	65.2	12.17	63.00	34.59
InfLLM	77.83	21.6	11.86	59.52	22.21
StreamingLLM	23.37	30.8	11.11	48.14	25.63
MemoryLLM†	16.55	13.6	9.01	30.00	16.11

Table 19: **Cross-benchmark performance of CoMem and external references.** The RULER column uses one common 15-cell task set. External rows differ from CoMem in interface, context treatment, adaptation, and sometimes backbone, so they are descriptive rather than matched comparisons. †MemoryLLM uses a released 7B model.

A.3 Full chat-template-free benchmark comparisons

Tables 20–24 give the full audited comparisons for RULER, LongEval, BABILong, and LongBench. Main rows use the unified no-chat protocol; they include KV-Direct (Qasim et al., 2026), InfLLM (Xiao et al., 2024a), StreamingLLM (Xiao et al., 2024b), and MemoryLLM (Wang et al., 2024). MemoryLLM uses its released Llama-3 checkpoint (Grattafiori et al., 2024), while the marked LLoCO row uses its released native setup (Tan et al., 2024). These are descriptive external references rather than matched rankings. The all-method LoCoMo comparison is Table 25; judge details are in Appendix B.

A.3.1 RULER and length extrapolation

Table 20 reports the complete common-support RULER grid; the matched positional-scaling comparison follows in Table 21. The matched positional-scaling grid is Table 21; the natural-task breakdowns begin in Tables 22–25.

A.4 Store scaling and failure boundaries

Table 26 tests the "bounded read over extensible store" claim directly. Retrieval is invariant when the required evidence fits top-12, but lookup and index construction still scale with store size. Beyond the budget, recall is physically capped; co-keyed multivalue evidence falls below even this cap (recall 0.62 at E=8), and a common-word-frequency task scores zero as coverage falls from 4.5% at 128k to 1.2% at 512k. These negative results delimit the claim: bounded retrieval is not global aggregation, and successful retrieval does not guarantee multi-hop reasoning.

We audited the PG-19 adapter corpus against the InfiniteBench long-book support using normalized 13-gram containment sketches. The conservative 0.8 threshold flags 24/86 books (163/580 records), but containment is sharply bimodal: 31 books are below 0.10, no book lies in 0.18–0.60, and 54 are at least 0.60. Any cut in 0.20–0.60 therefore flags 54/86 books and 387/580 records; anonymized character names make the 0.8 count an undercount. Existing predictions were unavailable for a strict-clean-subset rescore (113 QA and 76 choice records), so we remove the book-quality comparison rather than treat it as out-of-domain evidence. This does not affect the synthetic store-scaling results above. We subsequently audited the PG-19 adapter corpus against the natural-task suites. LoCoMo has no flagged overlap and LongEval is synthetic by construction. Within the six LongBench QA datasets, material overlap is localized to NarrativeQA: 96 of 200 examples are removed by the strict 13-gram clean filter, leaving 104. Recomputing all available rows changes six-task macro F1 by at most 0.13 points; for example, frozen CoMem changes 10.63→10.60 and KV-Direct 10.07→9.94. The PG-19-trained j=12 row changes 9.58→9.57. Thus the displayed LongBench ordering and near-ties are robust to this strict subset, although the audit detects near-verbatim overlap rather than semantic paraphrase or exposure through other pretraining corpora.

Method	Task	8k	16k	32k	64k	128k
CoMem + LoRA	single-2	100	100	99	99	100
	multikey	95	94	97	91	93
	var-track	96.6	97.6	98.8	99.0	95.8
CoMem frozen (j=9)	single	96	99	99	99	96
	multikey	44	44	59	30	48
	var-track	45	38.8	36.4	31.2	25.8
KV-Direct	single	100	100	100	100	0
	multikey	100	100	98	88	0
	var-track	100	100	99.8	96.2	0
InfLLM	single	100	100	92	61	57
	multikey	100	79	65	45	20
	var-track	100	96.8	90.6	81.8	79.2
StreamingLLM	single	85	36	20	10	2
	multikey	83	34	18	13	4
	var-track	41.8	2.2	0.6	0.2	0.8
External persistent-memory reference (different backbone)						
MemoryLLM†	single	29	40	37	21	21
	multikey	28	24	25	12	10
	var-track	0	1.2	0	0	0

Table 20: Full chat-template-free RULER comparison using official `string_match_all` (n=100 per cell). KV-Direct places the full context in the read with native RoPE and no YaRN; its 64k/128k cells are unextended stress references rather than a length-extended upper bound. CoMem keeps a bounded retrieved pack. This is **Cohort A**: every method uses `niah_single_2`, `niah_multikey_1`, and `variable_tracking`, with n=100 and the official scorer. Its CoMem macro is 97.05. Cohort B instead uses `niah_single_3` to match the YaRN arm (Table 21) and yields 96.07; the cohorts are never pooled in a direct comparison. †MemoryLLM is the principal external persistent-memory reference and uses its released Llama-3-8B-chat checkpoint; its different backbone precludes matched attribution.

Task	Method	8k	16k	32k	64k	128k
single needle	KV-Direct	100	100	100	100	0
	KV-Direct+YaRN†	100	100	100	100	**100**
	CoMem+LoRA	100	91	97	98	98
	CoMem+LoRA+YaRN†	100	90	97	97	98
multikey	KV-Direct	100	100	98	88	0
	KV-Direct+YaRN†	98	94	96	91	89
	CoMem+LoRA	94	91	99	90	**93**
	CoMem+LoRA+YaRN†	82	85	94	88	**93**
variable tracking	KV-Direct	100	100	99.8	96.2	0
	KV-Direct+YaRN†	99.2	99.4	26.6	67.2	57.8
	CoMem+LoRA	96.2	98.0	98.2	98.6	**99.0**
	CoMem+LoRA+YaRN†	88.6	90.0	89.4	95.0	93.6

Table 21: Matched-n=100 Qwen3-8B RULER scaling. †YaRN factor 4 gives an effective 163,840-token window. At 128k, unextended-backbone CoMem+LoRA scores 98/93/99.0 on single-needle, multikey, and variable tracking, versus 100/89/57.8 for KV-Direct+YaRN. Applying YaRN to CoMem changes these to 98/93/93.6, showing an in-window position-scaling tax on variable tracking. KV-Direct+YaRN is stronger on 128k single-needle and on several 8k–32k single/multikey cells; CoMem's advantage is concentrated in variable tracking and bounded read without full-prompt prefill.

Method	**8k**	**16k**	**32k**	**64k**	**128k**	**Mean**
Raw replay (j=0)	100	96	99	94	97	97.2
CoMem + LoRA	69	75	64	67	70	69.0
CoMem frozen (j=9)	8	0	4	0	4	3.2
KV-Direct	100	96	92	38	0	65.2
StreamingLLM	86	34	18	10	6	30.8
InfLLM	60	30	12	4	2	21.6
MemoryLLM†	22	22	16	6	2	13.6

Table 22: LongEval register-content accuracy. The raw-replay row reports the measured length-specific j=0 results (n=100/length), not the overview aggregate repeated across cells. Unmarked rows use the unified chat-template-free protocol. Means use the common 8k–128k support. KV-Direct uses native RoPE without YaRN, so its 64k/128k cells are unextended stress references and the five-length mean is not a length-extended upper bound. CoMem's separate six-length headline is 72.83 after including its 4k score of 92. The frozen CoMem control used 48 generation tokens rather than 16; both budgets exceed the single-number answer length. †MemoryLLM is the external persistent-memory reference and uses a different chat-tuned backbone, so its row is not a matched comparison.

Method	**Task**	**0k**	**1k**	**2k**	**4k**	**8k**	**16k**	**32k**	**Mean**
CoMem + LoRA	qa1	**98**	80	68	68	46	17	12	55.6
	qa2	26	44	43	44	23	8	1	27.0
	qa5	68	**76**	**76**	**75**	68	**60**	**58**	**68.7**
CoMem frozen (j=9)	qa1	**98**	59	71	57	7	1	4	42.4
	qa2	53	17	28	26	8	4	1	19.6
	qa5	70	74	65	60	41	39	40	55.6
KV-Direct	qa1	**98**	**84**	80	**74**	**80**	**72**	**63**	**78.7**
	qa2	**58**	53	**50**	46	**49**	**49**	**37**	**48.9**
	qa5	71	73	62	59	65	42	**58**	61.4
InfLLM	qa1	97	**84**	**81**	72	69	52	34	69.9
	qa2	51	**58**	41	47	41	39	30	43.9
	qa5	69	68	63	66	**73**	**60**	55	64.9
StreamingLLM	qa1	97	83	80	71	23	27	12	56.1
	qa2	49	56	44	**49**	19	11	4	33.1
	qa5	68	65	68	65	39	47	34	55.1
MemoryLLM†	qa1	52	46	35	28	23	17	12	30.4
	qa2	37	29	17	18	21	17	11	21.4
	qa5	50	45	39	35	35	34	29	38.1

Table 23: BABILong accuracy (n=100 per task–length cell), evaluated with the official `compare_answers` scorer and `TASK_LABELS`. Within the native context window, full-context KV-Direct is substantially stronger on qa1 and qa2, exposing the compression tax, whereas distilled CoMem achieves the highest qa5 mean, ahead of InfLLM (68.7 vs. 64.9). All methods are evaluated under the unified plain-text prompting protocol, with model-specific chat templating disabled (`chat_template=False`). †MemoryLLM is the principal external persistent-memory reference and uses its released Llama-3-8B-chat checkpoint. Its different backbone precludes matched attribution. We retain the uniform no-chat row and make no prompt-sensitivity claim without a paper-retained matched table.

Method	NQA	Qasper	Hotpot	2Wiki	MultiF.	Musique	Macro
Raw replay (j=0)	3.88	11.73	12.17	12.42	26.18	7.47	12.31
CoMem + LoRA	4.12	11.01	11.62	12.83	25.41	7.91	12.15
CoMem frozen (j=9)	4.63	11.23	9.41	10.71	22.05	5.72	10.63
KV-Direct	3.70	11.82	12.68	12.03	25.30	7.49	12.17
InfLLM	2.99	11.35	12.08	12.50	25.33	6.94	11.86
StreamingLLM	3.49	11.64	11.02	12.19	22.42	5.88	11.11
MemoryLLM†	3.13	8.46	8.76	10.27	17.43	5.98	9.01
LLoCO‡	24.21	24.45	44.24	—	—	—	30.97

Table 24: Official LongBench QA token F1 by dataset. The raw-replay row contains the measured per-dataset j=0 scores (n=150–200/dataset), with 12.31 as their six-task macro. Except for the explicitly marked LLoCO diagnostic, methods use the same plain-text no-chat protocol. CoMem with LoRA scores 12.15 and full-context KV-Direct scores 12.17; because only CoMem uses the split-interface adapter, this near-equality is descriptive rather than a training-budget-matched compression estimate. The frozen CoMem configuration reports adapter-free performance at j=9. NQA denotes NarrativeQA, and MultiF. denotes MultiFieldQA-en. †MemoryLLM is the principal external persistent-memory reference and uses its released Llama-3-8B-chat checkpoint. Its different backbone precludes matched attribution. We retain the uniform no-chat row and make no prompt-sensitivity claim without a paper-retained matched table. ‡LLoCO uses LLaMA-2-7B AutoCompressor, released per-domain supervised LoRAs, fp16, and its native task prompts. Released weights cover only NarrativeQA/Qasper/HotpotQA on our 200-example sets; 30.97 is their three-task macro and is not comparable to the six-task macro column.

Method	Judge	Judge$_{1:4}$	F1	Substr. acc.
Raw replay (j=0)	41.59	52.60	9.90	25.23
CoMem + LoRA (j=12)	38.27	48.64	9.15	23.36
KV-Direct	34.59	43.83	9.02	22.36
StreamingLLM	25.63	31.04	7.67	13.75
InfLLM	22.21	26.43	7.39	13.34
MemoryLLM†	16.11	20.65	5.91	9.52

Table 25: LoCoMo comparison on all 1,986 chat-template-free examples. The raw-replay row is the measured full-set j=0 reference rather than an overview aggregate duplicated into category cells. Judge$_{1:4}$ is GPT-4o accuracy on the 1,540 answerable items; the full Judge column folds in local abstention scoring for 446 adversarial items. F1 and substring accuracy are deterministic lexical diagnostics; semantic Judge is the primary metric. All rows use the same scoring protocol. The same-j adapter effect is isolated in Table 9. On paired answerable items, distilled CoMem exceeds KV-Direct by 4.81 points (item-bootstrap 95% CI $[2.34, 7.34]$); a 10-conversation cluster bootstrap gives $[1.27, 8.32]$ and 8/10 observed conversation differences favor CoMem. An independent DeepSeek-V3 audit preserves the ordering. †MemoryLLM is the external persistent-memory reference and uses its official chat-tuned Llama-3 checkpoint; its different backbone precludes matched attribution. The paper retains only its uniform no-chat evaluation and makes no native-chat sensitivity claim.

A.5 Reproducibility details

Tables 27–29 record the configuration, training, and evaluation settings used by the Qwen3-8B flagship. The archived configuration files and evaluation scripts accompany the code release.

Artifacts, licenses, and data scope. The study evaluates English text only. It uses the frozen released Qwen/Qwen3-8B checkpoint (Apache-2.0); we do no backbone continued pretraining. It also uses PG-19 (Rae et al., 2019), RULER (Hsieh et al., 2024), BABILong (Kuratov et al., 2024), LongBench (Bai et al., 2024), LongEval/LongChat (Li et al., 2023), and

Store	NIAH rec.	NIAH score	VT rec.	VT score
128k	1.00	100	1.00	18
256k	1.00	100	1.00	34
512k	1.00	100	1.00	40
1M	1.00	100	1.00	54
2M	1.00	90	1.00	38
4M	1.00	100	1.00	34

Store	Read tokens	BM25 (ms)
128k	6,211 / 6,463	80 / 20
256k	6,209 / 6,463	164 / 39
512k	6,209 / 6,464	328 / 81
1M	6,209 / 6,463	697 / 170
2M	6,211 / 6,463	1,407 / 357
4M	6,208 / 6,463	2,852 / 725

Table 26: Store scaling with a fixed top-12 read. Values separated by "/" are single-evidence NIAH / five-link variable tracking (VT). Evidence recall (n=20) stays perfect and the model read stays about 6.2–6.5k tokens from 128k to 4M stored tokens; BM25 lookup grows approximately linearly. Generation scores use n=10 and reveal that VT reasoning remains noisy even when all five evidence chunks are retrieved. At 128k, increasing required VT evidence from 12 to 16/24/32 gives recall 1.0 to 0.75/0.50/0.375, exactly $\min(1, 12/E)$.

Layer	Setting	Value
Backbone	Architecture	Frozen Qwen/Qwen3-8B; revision `b968826d9c46` (full hash in artifact manifest); L=36, d=4096, 32 query heads, 8 KV heads, bf16, SDPA.
Backbone	Positions	Native window 40,960; RoPE θ=10^6; no active RoPE scaling or YaRN.
Write	Cached state	Split j=12; disjoint 512-token chunks; chunk-local positions restart at zero; one bf16 residual tensor h_j per token.
Read	Pack	$[\text{BOS}; \text{selected } h_j; h_j(q)]$ in document order, with fresh contiguous RoPE and full causal cross-chunk attention. The nominal cap is $1 + 12\times512 + 512 = 6{,}657$ tokens; actual RULER means are about 6.2–6.5k because tail chunks and queries can be shorter.
Retrieval	BM25	Token-ID documents and bare-question token IDs; k_1=1.5, b=0.75, Robertson IDF with +1 smoothing; no lowercasing, stemming, or stop-word removal.
Retrieval	Iteration	`rounds=0` means automatic $\lceil k/h \rceil$ rounds, not one-shot. Each round uses newly selected chunks as the next frontier and deduplicates the selected set.
Retrieval	Width	All five benchmarks fix k=12, hop width h=4, and therefore three automatic rounds. Chunk size 512 and the BOS sink are also fixed.
Read	Mask	The pack is one causal sequence: the final query sees all chunks; later chunks may attend to earlier chunks. In the block-diagonal ablation, chunks see only themselves and the sink, while the final query still sees all chunks.
Generation	Decoding	Greedy decoding (no sampling, temperature, top-p, or beams); no chat template and no thinking mode.
Adapter-free arm	Deployment	Frozen backbone, no LoRA, j=9; all other memory and evaluation settings are held fixed.

Table 27: Flagship CoMem inference configuration. The retrieval and read configuration is frozen across all five benchmarks; only benchmark harness parameters differ.

LoCoMo (Maharana et al., 2024). RULER and LongChat code are Apache-2.0; the LongBench repository is MIT-licensed, while its component datasets retain their upstream terms; BABI-Long code is Apache-2.0 and incorporates BSD-licensed bAbI tasks; LoCoMo is CC BY-NC

Setting	Value
Adapter	Rank 32, α=64, dropout 0; Q/K/V/O and gate/up/down projections in layers 12–35 only; 58.20M trainable parameters (0.71%). Final weight SHA-256: `dd09cd17457c63578c0f38dab79b287ab5da6e3f14c119aedafec1c34400536f`.
Teacher and objective	The same Qwen3-8B with adapters disabled and j=0 serves as the frozen teacher. At every query token, the teacher top-64 vocabulary indices define a shared support; teacher and gathered student logits are each softmax-renormalized on that support, and outside-support logits are discarded. $\mathcal{L} = 0.6\,\mathrm{KL}(p\|q) + 0.4\,\mathrm{KL}(q\|p)$ at temperature 1, averaged over query tokens, with no CE term. Retained teacher top-64 probability mass was not logged.
Data	Plain PG19 train text; no synthetic long-context examples, retrieval, task labels, or benchmark supervision. Training uses 2,048-token windows, formed as four 512-token chunks.
Forward construction	Each 2,048-token window is split into three context chunks and one query chunk. Teacher and student use the same BOS sink, chunk token IDs, order, and query segment. The teacher replays every chunk from layer 0 with the adapter disabled; the student writes chunks independently through layers 0–11 without gradient and resumes one causal packed sequence through layers 12–35 with the LoRA. Loss is computed only on all query tokens.
Optimization	AdamW, $\beta = (0.9, 0.95)$, weight decay 0, peak LR 8×10^{-5}, 100-step linear warmup, cosine decay to zero, gradient clipping 1.0, seed 42, bf16, no gradient checkpointing.
Schedule	4,000 steps on 8 GPUs, one sample per GPU and no gradient accumulation (global batch 8), totaling about 65.5M tokens.
Compute	One node with 8×NVIDIA H20 GPUs. Logged throughput is about 24.5 samples/s, implying roughly 22 minutes wall-clock; training peak memory was not recorded.

Table 28: Self-distillation and LoRA training configuration. The adapter name's "4k" denotes 4,000 optimization steps; the training sequence length is 2,048 tokens.

Benchmark	Tasks and support	Generation	Scoring and aggregation
RULER	Single needle, multikey, and variable tracking; 8k–128k; 100 examples/cell; 8 shards; seed 42.	48 tokens; 60 for variable tracking.	Official case-insensitive `string_match_all`; equal-weight macro over 15 cells.
LongEval	Lines retrieval; 4k–128k; 100 examples/length; 8 shards; seed 1234.	16 tokens for the flagship; 48 for frozen controls.	First numeric answer exact match; six-length macro (five-length 8k–128k macro for matched baseline tables).
LongBench	Six QA datasets; 1,150 examples total; 8 shards; seed 42.	32/64/128 tokens by dataset.	Official multi-reference token F1; equal-weight macro over six datasets.
BABILong	qa1/qa2/qa5; 0k–32k; 100 examples/cell; 4 shards; seed 42.	20 tokens.	Official `compare_answers` with `TASK_LABELS`.
LoCoMo	All 1,986 QA pairs from 10 conversations; 4 shards.	48 tokens for the main 8B comparison; 512 for the cross-scale audit.	GPT-4o judge on categories 1–4; official local abstention rule for category 5; full-set judge is the headline.

Table 29: Benchmark-level evaluation settings. Every main comparison uses `chat_template=False`. MemoryLLM is the external persistent-memory reference, evaluated with its released chat-tuned backbone.

4.0. PG-19 contains pre-1919 Project Gutenberg books, for which the dataset page does not assert one uniform corpus license, so we do not redistribute its text. We likewise redistribute no model weights, benchmark examples, or private API responses. Subject to upstream

Length	Full time	CoMem time	Speedup	Full mem.	CoMem mem.
8k	0.165 s	0.309 s	0.53×	19.92 GB	17.60 GB
16k	0.356 s	0.441 s	0.81×	24.55 GB	17.67 GB
32k	0.826 s	0.681 s	1.21×	33.82 GB	17.79 GB
64k	2.104 s	1.200 s	1.75×	52.34 GB	18.04 GB
128k	6.035 s	2.202 s	**2.74**×	89.39 GB	**18.54 GB**

Table 30: Same-platform Qwen3-8B pipeline efficiency with the same distilled rank-32 LoRA attached to both paths (one NVIDIA L20A, 183 GB HBM, bf16/SDPA, median of three). This is a separate cohort from the stock-dense H20 measurement in Table 4. "Full" processes the entire context; "CoMem" includes document Write, CPU retrieval, query Write, and upper-layer Read over the selected pack, but excludes one-time BM25 index construction and external-store I/O. The select-first online path is reported separately in Table 4; it assumes a prewritten store and is not mixed with these full-pipeline measurements.

terms, the anonymous archive includes the adapter, source, documentation, pinned requirements, prediction hashes or permissible shards, judge prompts/parsed outputs, raw timing records, and third-party notices. Users obtain all external artifacts from their original providers and remain responsible for their terms. Table 29 reports the evaluated supports and sample counts; LoCoMo category denominators are in Appendix B.

Software environment. The archived environment specifies Python 3.10+, PyTorch 2.10, Transformers 5.5.4, PEFT 0.10+, Datasets 2.14+, bf16, and SDPA. Evaluation uses the benchmark-specific official scorers named in Table 29; the archive records all non-default flags and provides a CPU correctness gate.

Compute accounting. The final adapter run used one node with 8 H20 GPUs for about 22 minutes, or approximately 2.9 H20 GPU-hours. Table 28 and this section report model size, trainable parameters, hardware, and measured timing protocol. Total GPU-hours across preliminary probes, failed runs, baseline generation, and all ablations were not consistently logged; we report this omission rather than extrapolating an unreliable total.

A.6 Efficiency, replay attribution, and robustness

Efficiency measurement. The Write-inclusive cohort uses one NVIDIA L20A (183 GB HBM) with PyTorch 2.10, Transformers 5.5.4, bf16, SDPA, and the rank-32 LoRA enabled, reporting medians of three repetitions. Full-context prefill times the complete forward; CoMem times document Write, CPU retrieval, query Write, and upper-layer Read through logits. The timer excludes one-time BM25 index construction and external-store I/O; Table 34 measures the latter separately. At 128k, full-context prefill is 6.035 s and peaks at 89.39 GB, while CoMem takes 2.202 s and peaks at 18.54 GB. Cached decode uses 20 greedy steps.

The store-ready online-prefill cohort above is the paper's only dense-context online reference. The dense arm is stock Qwen3-8B without the adapter, while CoMem uses the flagship LoRA, so it is a composed deployment point rather than an adapter-matched depth estimate. The same-adapter L20A pipeline in Table 30 is the complementary Write-inclusive reference.

The main serving experiment instead measures write-once/fetch/reuse directly. At 32k, CPU-pinned break-even is 8.9–10.9 queries for generations up to 128 tokens, while the complete GPU/CPU grid is 5.5–10.9. At 128k, the $G \leq 128$ crossover spans 25.8–37.4 queries; at 1M it spans about 180–421. The difference from the earlier component cohort comes from source length, generation length, storage tier, and timing boundary. CoMem stores 8 KiB/token (about 1 GiB at 128k), far more than text but 18× less than full KV. Main timing cohorts keep the store in HBM; Table 34 separately measures placement beyond HBM, so its I/O microbenchmark is not added to model latency.

(a) Quality at a 6,657-token/KV budget

Method	Native A macro	Full-prompt 64k/128k single	multikey	tracking	LoCoMo F1/acc
CoMem+LoRA	**97.05**	99/100	91/93	99.0/95.8	9.15/23.36
SnapKV	72.33†	100/100	94/91	80.0/84.8	9.21/22.05
PyramidKV	72.32†	100/100	93/89	88.6/86.8	9.13/22.10

(b) Full-prefill systems cost on one H20

Method	Len.	ms	GB	ms/tok
SnapKV	8k	1,213	18.6	24.7
	32k	6,342	25.7	24.6
	128k	51,520	54.3	24.7
PyramidKV	8k	1,203	18.6	24.7
	32k	6,301	25.7	26.7
	128k	51,520	54.3	26.8

Table 31: Selected/compressed-KV baselines at a similar retained-KV size but a different full-prefill boundary. Panel (a) uses RULER Cohort A and full-set LoCoMo lexical scoring. CoMem reads its bounded pack from the full store without YaRN; the SnapKV/PyramidKV 64k/128k columns use YaRN factor 4 and see the full prompt. †Their native 15-cell macros include 64k/128k inputs left-truncated to the 40,960-token native window; within 8k–32k both score 99.96. Thus 72.33/72.32 are not common-native-support quality summaries. Panel (b) times full-prompt prefill, compression, and 64-token decode (median of three); the final column is decode latency. These methods must prefill the full prompt before eviction. CoMem's separate L20A cohort peaks at 18.54 GB at 128k (Appendix A.6); hardware differs, so we report no cross-table latency ratio.

Retained-KV references. Table 31 reports SnapKV and PyramidKV (Li et al., 2024; Cai et al., 2025) under their full-prefill boundary. They answer a different systems question from cross-query persistence and are not treated as nearest modular-cache baselines.

Same-backbone full-depth chunk-KV baseline. Table 6 uses a minimal faithful CacheBlend-style implementation on Qwen3-8B: full-depth per-chunk KV, exact global RoPE reindexing, and selective context-token recomputation. The fp32 real-model self-test gives maximum KV reindex deviations below 4.4×10^{-5} and makes $r{=}1$ agree with full prefill to 2.8×10^{-5} maximum logit error with 100% top-1 agreement. The released aggregate contains all 48 RULER ratio/task/length cells, 24 BABILong cells, and four full-set LoCoMo ratio points with no missing required cells. Full-depth Qwen3 GQA KV is $2\cdot36\cdot8\cdot128\cdot2=147{,}456$ bytes/token, versus 8,192 bytes/token for one bf16 residual. The baseline is training-free and does not include CacheBlend's complete native scheduler or cache manager.

Paired quality–latency replay control. Table 32 uses a separate H20 harness. The two arms share the Qwen3-8B checkpoint, flagship LoRA SHA-256 and all 168 mounted modules, example IDs, decode parameters, and per-example retrieved chunk IDs, order, and pack tokens; only replay start differs. Across 15 RULER Cohort-B cells ($n{=}100$ each), full replay scores 99.19 and layer-12 replay scores 96.07. The paired 3.12-point gap has bootstrap 95% CI $[2.36, 3.93]$ and exact McNemar $p{=}8.79\times10^{-24}$ (83 full-only correct versus one layer-12-only correct). The gap is largest on distractor-heavy multikey retrieval.

For latency, each arm uses the same approximately 6.5k-token pack retrieved from 16k source contexts in three independent processes, with three warmups and 20 reads per process. Layer-12 replay is 1.403× faster (931.9 to 664.4 ms), with all three process-level ratios between 1.402 and 1.404. Total-decode medians remain similar at roughly 2.76–2.86 s; adding the observed Read and decode components gives only about 1.07–1.09× total ratios. A preceding component harness attributes the read reduction primarily to upper-transformer forward (1.398×), with unchanged LM-head time. Retrieval, index construction, reusable store/query Write, and persistent I/O are outside this read-phase timing. The result is therefore a measured quality–latency trade-off, not quality-preserving or end-to-end acceleration.

Replay start	RULER	Read (ms)	p10	p90
j=0	**99.19**	931.9	931.6	942.0
j=12	96.07	**664.4**	663.8	667.1
Trade-off	−3.12 pt	1.403× speedup		

Table 32: Paired quality–latency control with the same LoRA. Quality is the 15-cell RULER macro (n=1,500 paired examples); the 3.12-point loss has paired-bootstrap 95% CI [2.36, 3.93]. Latency uses a fixed approximately 6.5k-token pack retrieved from 16k source contexts and reports medians across three independent processes (20 reads each). Retrieval, reusable Write, and I/O are excluded. Total decode is similar across arms (about 2.76–2.86 s), reducing the combined Read-plus-decode ratio to roughly 1.07–1.09×.

Replay pathway	RULER B	MK 8/16k	Reusable
Full replay from j=0	99.19	100.0	No
Continuous-prefix h_{12} oracle	99.19	100.0	No
Chunk-local cached h_{12}	96.07	92.5	Yes

Table 33: Write-side attribution with identical examples, packs, and LoRA. The continuous-prefix oracle runs layers 0–11 over the selected packed sequence for every query, then starts upper replay from that h_{12}; it is bit-identical to full replay on all 1,500 Cohort-B examples. The deployable chunk-local path loses 3.12 points (95% CI [2.36, 3.93]; McNemar p=8.79×10^{-24}), and 7.50 points on the pooled 8k/16k multikey cells (n=200, CI [4.0, 11.5]). The oracle is an attribution upper bound, not a reusable cache.

Backend	Write GB/s	Fetch ms	H2D ms	Peak QPS	16M outcome
GPU resident	OOM	–	–	–	128 GiB > 95 GiB
CPU pinned	139.6	0.81	1.41	941	235 GB host
Local NVMe	2.45	13.42	0.91	264	2.18 GB host
Network/CEPH	1.66	79.20	1.20	42	2.19 GB host

Table 34: Measured persistent-store tiers at 16M tokens (128 GiB of bf16 h_{12}). Each query fetches the same 50.3 MB top-12 pack; medians use seven repetitions after two warmups, and QPS is the best of 1/4/16 concurrent clients. GPU residence fits 8M tokens (64.98 GB peak) but not 16M; off-GPU tiers retain store-size-independent transfer volume at increasing I/O latency. This microbenchmark times storage and transfer, not BM25 or model inference.

A.7 Seed-plus-effective-batch robustness

Table 35 reports the available run-level aggregates. Across the 18 individual RULER and BABILong cells, the median cell-wise SD is 1.34 points and the maximum is 4.36. Because the two added runs change effective batch from 8 to 3, these numbers jointly probe initialization and optimization-noise robustness; they are not a clean three-seed variance estimate. The available evidence also does not support a multi-run claim for the exact 15-cell RULER-B, LoCoMo, or LongEval headlines.

A.8 Prompt and native-chat sensitivity

For Qwen3-8B, the GPT-4o LoCoMo judge is substantially less prompt-sensitive than token F1. CoMem scores 38.27 without chat templating and 37.76 with it; KV-Direct scores

Suite	Adapter	Seed 42	Seed 1	Seed 2	Mean	SD
RULER 9-cell macro	batch 8 / 3 / 3	98.80	99.42	98.76	98.99	0.37
BABILong 9-cell macro	batch 8 / 3 / 3	38.44	40.33	41.44	40.07	1.52
RULER multikey 3-length mean	batch 8 / 3 / 3	96.67	98.67	96.67	97.33	1.15
BABILong qa5 3-length mean	batch 8 / 3 / 3	66.00	69.33	71.00	68.78	2.55

Table 35: **Seed-plus-effective-batch robustness, not clean seed variance.** All adapters use the same Qwen3-8B split, PG-19 data budget, and rank-32 objective; seed 42 is the flagship effective-batch-8 run, while seeds 1 and 2 use effective batch 3 with matched examples seen. Evaluation uses identical cells and supports: RULER single/multikey/variable-tracking at 32k/64k/128k (n=50/cell) and BABILong qa1/qa2/qa5 at 4k/16k/32k (n=100/cell). SD is the sample SD across the three run-level cell macros. These reduced-support cells are not the 15-cell RULER-B headline, and no LoCoMo or LongEval multi-run aggregate was retained.

34.59 and 38.22. In contrast, CoMem's F1/substring accuracy changes from 9.15/23.36 to 19.51/28.65, while KV-Direct F1 changes from 9.02 to 40.06, showing that lexical metrics are strongly formatting-sensitive. We therefore retain the uniform no-chat judge comparison as the controlled headline.

For MemoryLLM, we report only the uniform no-chat row in the paper. A matched native-chat/no-chat table was not retained in the frozen manuscript; therefore we make no prompt-sensitivity claim for this external baseline and do not mix prompt protocols within an average.

A.9 Cross-scale and MoE generality

Adapter-free sweeps across six Qwen3 sizes and Qwen3-30B-A3B confirm that the split-forward implementation ports beyond the flagship. Performance is non-monotonic because split, architecture, and task differ, so we infer neither a scaling law nor a replicated quality frontier. These exploratory values and raw predictions are retained in the artifact; the Hy3 study below gives the stronger architecture-level partition check reported in this appendix.

A.9.1 Hunyuan Hy3: an 80-layer sparse MoE

We additionally port CoMem to Tencent Hunyuan Hy3 (Tencent Hunyuan Team, 2026), an 80-layer sparse Mixture-of-Experts model (192 experts, top-8 routing plus one shared expert; $\approx$ 597 GB in bf16) with a 262,144-token native window. The backbone is sharded across eight GPUs; Write and Read semantics are unchanged.

Exact partition through MoE routing. Reconstructing a contiguous forward by writing `layers[0:j]` and resuming `layers[j:80]` gives $\max|\Delta\text{logit}| = 0.0$ at $j \in \{0, 1, 40, 80\}$, including the Dense-to-MoE boundary. The resumed half re-executes every router and expert, so the partition preserves routing exactly on this self-test.

Split depth and self-distillation. A PG19 sweep places the best readout region near $j \approx 32$–36. At $j = 32$, self-distillation reduces the 16k multiplicative LM tax from 1.496 to 1.078 and raises top-1 agreement from 0.773 to 0.871 (Table 36).

Bounded read through 256k. With the distilled adapter and BM25 top-8, the read stays at $\approx$ 4.3–4.6k tokens from 16k through 256k. CoMem reaches 100/98 single/multikey recall at the 256k native ceiling with no OOM (Table 37).

	ctx 8k		ctx 16k	
j=32	gap	top1	gap	top1
zero-shot	1.378	0.823	1.496	0.773
distilled	**1.071**	**0.895**	**1.078**	**0.871**

Table 36: CoMem self-distillation on Hy3 at split j=32 (LoRA r=32, α=64 on `layers[32:]`, teacher j=0 RAG recompute, 4000 steps of plain PG19 KL, no synthetic data). **gap** is the multiplicative LM tax (CoMem-readout perplexity / full-context perplexity; 1.0 is exact); **top1** is argmax agreement with the full forward, over matched 16-doc PG19 windows. The 16k LM tax falls from +49.6% to +7.8% and top1 rises 0.773 $\rightarrow$ 0.871, reproducing the 8B-backbone self-distillation effect on the 80-layer MoE.

Task	16k	32k	64k	128k	256k
niah_single_2	98	92	100	98	**100**
niah_multikey_1	100	94	100	100	**98**
read length (tok)	4.3k	4.6k	4.5k	4.4k	4.5k
context (tok)	16k	32k	65k	131k	261k

Table 37: CoMem on the Tencent Hunyuan **Hy3** backbone (`hy_v3`, 80-layer sparse MoE), RULER needle recall (n=50, official `string_match_all`), split j=32, BM25 top-8, self-distilled LoRA. The read length fed to the model stays ≈4.3–4.6k tokens while the context grows 16× (16k→256k); CoMem retains 100/98 recall at the backbone's 256k native ceiling, with 0 out-of-memory failures. The context row reports the mean packed length of the RULER samples.

B Chat-Template-Free Statistical Verification

All statistics in this appendix were recomputed from the saved prediction shards on the shared result filesystem, without GPU reruns. The Qwen3-8B flagship uses the self-distilled j=12 adapter, chunk 512, top-12 retrieval, a BOS sink, and no chat template. The released adapter artifact is identified by the SHA-256 and configuration in Table 28. The main RULER directory uses the `iter_bm25` implementation; the saved configuration records `rounds=0` and an observed read of approximately 6.5k tokens. Every reported RULER cell contains eight of eight shards, 100 examples, no empty prediction, and zero mismatch between the stored score and a fresh official-kernel recomputation. BABILong likewise contains 100 valid examples in every cell.

B.1 RULER cohort definitions

Task	**8k**	**16k**	**32k**	**64k**	**128k**
niah_single_3	100.0	91.0	97.0	98.0	98.0
niah_multikey_1	94.0	91.0	99.0	90.0	93.0
variable_tracking	96.2	98.0	98.2	98.6	99.0

Table 38: RULER **Cohort B**, used for the matched length-extension and j=0 comparisons. The n=100 macro is 1441.0/15 = 96.0667; the exact single-needle variant is `niah_single_3`.

Cohort A is the all-method table (Appendix Table 20): it substitutes `niah_single_2`, keeps the same multikey/tracking tasks and n=100, and gives CoMem 97.05. Cohort B above was freshly sampled with `niah_single_3` because that is the exact task available for both YaRN arms and j=0; it gives 96.07. Tables mark every RULER aggregate with A or B, never subtract across them, and exclude legacy 256k cells. Table 21 contains only Cohort B.

B.2 LoCoMo judge protocol and denominator

Prediction shards for all methods and the adapter-free CoMem control deduplicate to all 1,986 LoCoMo items. GPT-4o judges the 1,540 answerable items in categories 1–4. Category 5 contains 446 adversarial abstention items and is scored locally: an answer is correct iff it is empty or matches the refusal rule. Thus the canonical full-set judge is over all 1,986 items, not only the API-judged subset.

Method	Cat. 1	Cat. 2	Cat. 3	Cat. 4	Cat. 5	Overall
CoMem + LoRA	**26.95**	**19.00**	**30.21**	**69.32**	2.47	**38.27**
CoMem frozen (j=9)	18.79	13.40	27.08	53.75	1.12	29.15
KV-Direct	24.11	18.69	25.00	62.19	2.69	34.59
StreamingLLM	22.70	11.21	23.96	42.21	6.95	25.63
InfLLM	18.44	14.33	25.00	33.89	**7.62**	22.21
MemoryLLM	15.60	8.72	15.63	27.47	0.45	16.11

Table 39: LoCoMo GPT-4o judge by category. Categories 1–4 contain 282/321/96/841 judged items; Category 5 contains 446 locally scored abstention items. Overall aggregates all 1,986 examples.

The judge was evaluated on July 22, 2026 using an OpenAI-compatible chat-completions endpoint with model name `gpt-4o`, `seed=1`, and no client-set temperature or top-p. The endpoint does not expose a dated model snapshot. Saved item-level records contain the item ID, binary parsed decision, raw response text, and model string; the anonymous artifact includes the permissible parsed decisions and their hashes rather than API credentials. Requests are retried four times with exponential backoff; unparsable responses and API failures are conservatively scored wrong. The prompt presents the question, gold answer, and prediction; it accepts any semantic match (including equivalent date phrasing), rejects contradictions, omissions, empty answers, and refusals, and requires exactly `CORRECT` or `WRONG`. The verbatim template is released with the evaluation script. Gold alternatives are joined with "OR." A reply must begin with `CORRECT` or `WRONG`; a fallback substring vote is used before defaulting to wrong.

On the common GPT-4o-judged items (n=1540), CoMem scores 48.64 and KV-Direct 43.83. A paired per-item bootstrap of their difference (10,000 resamples, seed 1234) gives +4.81 points with 95% CI $[2.34, 7.34]$. Because the questions are nested within only 10 conversations, this interval is not our sole inferential check. A paired conversation-cluster bootstrap, resampling the 10 conversations rather than individual questions (same 10,000 resamples, seed 1234), gives the same +4.81 point estimate with 95% CI $[1.27, 8.32]$ — wider than the per-item interval, as expected with only ten clusters; 8 of the 10 observed conversation-level differences favor CoMem. We do not interpret the bootstrap tail fraction as a calibrated hypothesis-test p value. We report the latter as a dependence-aware robustness check rather than claiming that 1,540 questions are independent. The canonical full-set point scores are 38.27 and 34.59, respectively.

B.3 Independent-judge audit

On July 26, 2026, we reran the binary prompt with DeepSeek-V3 on a stratified 200-item subset shared by CoMem and KV-Direct. Agreement with GPT-4o is 0.81 ($\kappa = 0.626$). Both judges preserve the ordering: GPT-4o scores 36.5/32.5 (+4.0) and DeepSeek-V3 56.0/49.0 (+7.0) for CoMem/KV-Direct. Absolute calibration differs, but the positive gap is judge-robust.

B.4 Uncertainty and aggregation checks

Equal-latency dependence reanalysis. We reanalyzed the saved anchor scores without model execution. The estimand is the equal-weight mean of the nine cell mean paired differences CoMem(k=12) minus replay(k=10). A *stratified fixed-cell* bootstrap keeps all nine cells and resamples exactly 100 paired examples within each cell, giving 95% intervals

$[-14.33, -8.78]$ for BM25 and $[-4.56, 2.56]$ for BGE. A *hierarchical* bootstrap first resamples nine cell labels with replacement, then independently resamples 100 paired examples inside each selected occurrence; its intervals are $[-18.67, -5.11]$ and $[-10.67, 8.33]$. Dropping one cell at a time gives ranges $[-13.13, -9.50]$ and $[-3.50, 1.75]$, respectively. The original pooled-IID intervals, $[-14.44, -8.67]$ and $[-4.67, 2.67]$, are retained only as sensitivity. All use 100,000 replicates and deterministic base seed 20260804. The first 100 LoCoMo examples are all conversation 0, leaving one observed cluster; a conversation bootstrap is therefore not identifiable. The release includes the score-only exports (without benchmark text, answers, or predictions), script, result JSON, and SHA-256 manifests.

Per-method, 1,000-resample item-bootstrap intervals (seed 1234) are $[36.20, 40.58]$ for CoMem's full-set LoCoMo judge, $[32.48, 36.71]$ for KV-Direct's, $[46.04, 50.98]$ and $[41.23, 46.30]$ for their category-1–4 judge scores, and $[8.60, 9.72]$ and $[8.51, 9.61]$ for their F1 scores. These unpaired, item-level intervals are descriptive; the paired cluster analysis above is the appropriate dependence-aware comparison.

We do not reuse cell-bootstrap intervals from the superseded RULER cohort or legacy BABI-Long predictions. The separate six-length LongEval aggregate is 72.83 over 600 examples (95% CI $[69.33, 76.33]$); the benchmark summary uses the predefined five-length 8k–128k mean of 69.0.